

A smart glove for Parkinson's severity detection via low-frequency motion features and on-device machine learning

Priyam Parikh¹ , Vidhi Srivastava² , Keyur D. Joshi³ , Mohd Asif Shah⁴ and Ankit Sharma⁵

Abstract

Parkinson's disease (PD) is a progressive neurological disorder affecting fine motor control, yet current assessments rely heavily on brief clinical observations, lacking continuous and objective measurement tools. This study aims to develop a low-cost, real-time system for classifying PD severity using motion data captured by a wearable smart glove. We address three key limitations in existing research: the absence of context-aware monitoring during daily activities, limited use of low-frequency tremor analysis for severity detection, and the lack of embedded models that report system-level performance metrics such as inference time and memory usage. Our glove integrates an Arduino Nano BLE microcontroller and BMI270 IMU sensor to record triaxial hand motion during resting, spoon holding, and eating tasks. Using a 4-second sliding window, we extract a compact set of time-domain (RMS, standard deviation, SMA, ZCR) and frequency-domain (dominant frequency, bandpower) features. These are used to train four lightweight machine learning models – Decision Tree, Logistic Regression, Naïve Bayes, and a Shallow Neural Network – evaluated via subject-independent validation. The Naïve Bayes model achieved the best trade-off with 93.67% test accuracy and real-time inference capability on-device. Our contributions include a reproducible embedded pipeline for PD assessment, integration of quantized models via TensorFlow Lite Micro, SHAP-based interpretability for model transparency, and benchmarking against recent studies. This work demonstrates the feasibility of deploying interpretable, real-time AI models on microcontrollers for practical PD severity monitoring in home or clinical environments.

Keywords

Parkinson's disease assessment, tiny machine learning algorithms, embedded machine learning, Fast Fourier Transform, smart glove

Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder that affects millions of individuals worldwide, significantly impairing motor function and quality of life. Tremors, rigidity, and bradykinesia are hallmark symptoms that manifest predominantly in the upper limbs, leading to challenges in performing daily activities such as holding utensils or self-feeding. Traditional clinical assessments of PD severity, including tools such as the Unified Parkinson's Disease Rating Scale (UPDRS),¹ are based on episodic, in-clinic evaluations that are inherently subjective, time-constrained, and dependent on physician expertise.² As a result, these assessments fail to capture fluctuations in symptom severity that occur throughout

¹Product Design Department, School of Design, Anant National University, Ahmedabad, India

²Product and Interaction Design Department, Institute of Design, Nirma University, Ahmedabad, India

³School of Engineering and Applied Science, Ahmedabad University, Ahmedabad, India

⁴STEM Department, Kardan University, Kabul, Afghanistan

⁵Centre for Research Impact & Outcome, Chitkara University Institute of Engineering and Technology, Punjab, India

Corresponding author:

Mohd Asif Shah, STEM Department, Kardan University, Parwan e Du, 1001, Kabul, Afghanistan.

Email: m.asif@kardan.edu.af



Creative Commons Non Commercial CC BY-NC: This article is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access pages (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

the day, particularly in fine-motor tasks performed during real-world activities. Consequently, there is a critical need for low-cost, objective, and continuous monitoring systems that can quantify PD severity in real time using unobtrusive wearable technologies. The goal of this study is to design and validate a smart glove-based system that leverages embedded machine learning and low-frequency motion features to assess PD severity in real time. The system integrates a lightweight sensor architecture, a compact feature extraction pipeline, and on-device inference capabilities, making it suitable for both clinical and home-based monitoring. This goal is motivated by the increasing demand for scalable digital health solutions that can bridge the gap between clinical evaluations and real-world symptom monitoring. To pursue this goal, we identify the following key research problems that have remained inadequately addressed in prior literature:

1. **Limited context-awareness in existing assessments:** Most prior works evaluate PD symptoms in controlled laboratory settings or during isolated clinical tests, which do not reflect the variability and context of symptoms during daily life tasks such as eating or resting. This limits ecological validity.
2. **Under-utilization of low-frequency motion features:** Fine-motor tremors associated with early and moderate PD stages often manifest in the 1–6 Hz range. However, many systems focus on high-frequency, gross-motor signals or overlook frequency-domain features entirely.
3. **Dependence on high-end computation:** Several existing approaches rely on cloud-based inference, smartphones, or laptops, which hinders real-time analysis and raises concerns regarding data privacy and latency.
4. **Lack of embedded performance metrics:** Studies that deploy machine learning models on wearables often report only accuracy but fail to report key system-level metrics such as inference latency, memory footprint, and real-time feasibility on microcontrollers.
5. **Absence of replicable, end-to-end pipelines:** Despite the growing availability of embedded machine learning toolkits, there is a lack of well-documented, reproducible pipelines that take raw sensor data through feature extraction, model selection, embedded deployment, and interpretability.

This restriction highlights the importance of continuous, unbiased, and real-time monitoring solutions. Advances in wearable technology have made new solutions for PD control a reality. Smart gloves with inertial measurement units (IMUs) and other sensor systems have come to be seen as potential solutions for recording precise hand and finger movement. These sensors allow tremor and other motor symptoms to be monitored in naturalistic conditions and give clinicians and investigators useful information.³ For instance, a group of researchers introduced a smart glove integrating flexible piezoelectric and inertial sensors, demonstrating its efficacy in evaluating PD-related hand movements and tremors.¹ The integration of machine learning (ML) algorithms with wearable devices has further enhanced the potential for real-time PD assessment. ML models can analyze complex sensor data to classify the severity of motor symptoms, offering personalized insights into disease progression. Recent research has explored various ML techniques, including decision trees, logistic regression, and neural networks, for tremor classification and PD severity assessment.^{4,5} To make it ready for ML algorithms. It is also important to measure continuous FFT as a training data of ML models. It is commonly employed to convert time-domain acceleration signals into the frequency domain, enabling the extraction of features pertinent to tremor analysis. By analyzing the frequency components of hand movements, FFT aids in distinguishing between different types and severities of tremors.⁶ This approach has been instrumental in enhancing the accuracy of ML models in PD assessment.^{4,5} We introduce in this context a smart glove system based on an Arduino Nano BLE microcontroller to monitor PD patient hand tremors while resting, grasping a spoon, and eating. The triaxial acceleration signals are recorded by the glove and processed with FFT for frequency-domain features. These features are evaluated by embedded ML models decision tree, logistic regression, Naïve Bayes, and a Shallow neural network to identify in real-time the PD severity. We apply a dataset with 150 participants with and without PD with different severities. We assess the ability of each ML model in accuracy, inference latency, and resource usage in order to determine the best method for real-time wearable PD evaluation. This work advances the body of wearable health sensing by presenting the possibility of embedding ML algorithms into sensorized gloves for continuous evaluation of Parkinson's disease. Our results have the potential to shape the design of affordable and non-invasive solutions for PD treatment and contribute to enhanced patient care via timely and targeted interventions.³ These models, when deployed on embedded systems like the Arduino Nano BLE, facilitate on-device processing, reducing the need for continuous data transmission and preserving patient privacy.²

This opportunity creates the need for a new approach that combines embedded intelligence with interpretability, efficiency, and ease of replication. Our study builds upon these identified gaps by designing a smart glove system that integrates an Arduino Nano BLE 33 microcontroller with a BMI270 inertial motion unit (IMU). The glove is designed to collect tri-axial acceleration data during three everyday tasks: resting, holding a spoon, and bringing the spoon to the mouth. These tasks are chosen deliberately to reflect fine-motor activity across different stages of PD. The recorded signals are segmented using a sliding 4-second window and passed through a lightweight feature extraction pipeline. We compute both

time-domain features Root Mean Square (RMS), Standard Deviation (STD), Zero Crossing Rate (ZCR), Signal Magnitude Area (SMA), and frequency-domain features using Fast Fourier Transform (FFT), including Dominant Frequency and Bandpower. These features are compact enough to run on resource-constrained devices while preserving the discriminatory power required for PD severity classification. We evaluate four lightweight machine learning models: Decision Tree, Logistic Regression, Naïve Bayes, and a Shallow Neural Network, all trained using subject-independent validation to avoid data leakage and ensure generalization. The Naïve Bayes model, in particular, achieves the highest test accuracy of 93.67%, while also maintaining low inference latency and minimal memory usage. This balance makes it suitable for real-time, on-device deployment using TensorFlow Lite Micro. Beyond the technical development of the glove, this study contributes to the scientific body of knowledge in several ways:

- It demonstrates that low-frequency motion features (typically below 6 Hz), often overlooked in prior PD studies, can be highly informative for assessing tremor severity in fine-motor tasks.
- It presents a resource-aware embedded learning pipeline that includes quantization, memory profiling, and inference time measurements, which are crucial for real-world deployment but rarely reported in previous literature.
- The study adopts a subject-independent evaluation protocol, which is essential for assessing model generalizability across users and ensures robustness beyond user-specific calibration.
- We also integrate SHAP-based interpretability, allowing clinicians and developers to understand which features drive the classification, thus promoting transparency in medical AI systems.
- Our findings are benchmarked against recent state-of-the-art systems and demonstrate comparable or superior performance using a significantly more compact and deployable architecture.

In addition to its scientific contributions, the system is designed to be cost-effective, reproducible, and open for extension. The use of Arduino Nano BLE and TensorFlow Lite Micro enables portability and encourages adoption in low-resource settings. The glove is lightweight and non-invasive, suitable for daily wear by elderly patients, and the end-to-end workflow supports real-time inference without requiring external computation. The novelty of this study does not lie in the use of any single component (e.g., accelerometer, FFT, or Naïve Bayes model), but in the cohesive integration of sensing, processing, and decision-making within a wearable platform that is fully self-contained and interpretable. By providing a complete pipeline from signal acquisition to model deployment, this work serves as a replicable blueprint for future researchers in wearable healthcare and edge AI. This study addresses critical gaps in Parkinson's monitoring by designing a smart glove that captures fine-motor activity during natural tasks, extracts lightweight yet informative features, and classifies PD severity using efficient embedded models. Our approach combines accuracy with interpretability and embedded feasibility, paving the way for real-time, at-home monitoring solutions for people living with PD. The findings contribute both practical solutions and replicable scientific methods for the advancement of embedded healthcare systems.

The next section presents the detailed design of the smart glove system and the objectives that shaped its development. This is followed by a description of the dataset acquisition protocol and signal segmentation techniques used to collect and prepare the triaxial motion data. The subsequent section outlines the feature extraction process, where both time-domain and frequency-domain features are computed. Thereafter, we discuss the selection and training of four lightweight machine learning models, along with the deployment process on the Arduino Nano BLE using TensorFlow Lite Micro. The paper then presents comprehensive results, including classification accuracy, inference latency, memory usage, and SHAP-based interpretability. A comparative benchmarking with recent studies is included to validate our approach. Finally, the paper concludes with a detailed discussion of the scientific contributions, system limitations, and future directions for scaling and clinical integration.

Related work

Smart gloves have emerged as promising tools for capturing fine-grained hand movement data. A group of researchers¹ have developed a smart glove embedded with flexible piezoelectric and inertial sensors capable of evaluating Parkinsonian tremors during different motor tasks. The glove demonstrated high precision in motion tracking and tremor analysis, making it a viable tool for non-invasive assessment. Li et al.⁷ introduced a multimodal sensor glove integrating pressure and motion sensors to monitor and classify Parkinson's hand movements. Using machine learning models such as K-Means and Back Propagation Neural Networks, the system achieved 95.83% classification accuracy across various hand postures. In a broader context, Del Din et al.⁸ reviewed the application of wearable sensor technologies in PD, including wrist-worn and body-mounted IMUs. Their findings emphasized the growing clinical relevance of wearable, especially for remote monitoring of gait, balance, and tremors. Sapienza et al.⁹ conducted a systematic review evaluating the clinical utility of inertial sensors for home-based PD monitoring. They noted a gap between technical validation and clinical implementation,

calling for standardized evaluation methods in real-life use cases. Finally, the iTex Glove developed by Ravichandran et al.¹⁰ provides an e-textile-based system for remote assessment of PD symptoms. This glove enables in-home tele-monitoring, capturing both voluntary and involuntary hand movements through embedded sensors, and has shown reliability in early-stage telemedicine trials. These studies collectively validate the potential of glove-based and wearable sensor systems in the accurate monitoring of PD symptoms, with a growing trend toward embedded, real-time, and patient-centric solutions.

Accurate analysis of tremor signals is essential in the assessment and monitoring of Parkinson's disease (PD). Various signal processing techniques have been employed to extract meaningful features from raw sensor data, facilitating the classification and severity estimation of tremors. Fast Fourier Transform (FFT): FFT is a widely used technique that transforms time-domain signals into the frequency domain, enabling the identification of dominant tremor frequencies. Balas et al.¹¹ utilized FFT in their tremor measurement system, effectively distinguishing between PD and other neurological disorders by analyzing frequency components of tremor signals. Wavelet Transform: To address the non-stationary nature of tremor signals, wavelet transform provides a time–frequency representation, capturing both temporal and spectral information. Wagner et al. proposed a wavelet-based approach for monitoring PD symptoms, demonstrating its efficacy in detecting tremor characteristics from wrist-worn sensors.^{12,13} Power Spectral Density (PSD): PSD analysis quantifies the power distribution of a signal over frequency, offering insights into tremor intensity and frequency. In their study, researchers applied PSD to differentiate between rest, postural, and kinetic tremors, enhancing the understanding of tremor dynamics in PD patients.¹⁴ Resampling Techniques with Machine Learning: Combining signal processing with machine learning, Almahadin et al.¹⁴ enhanced tremor severity classification by integrating resampling techniques with classifiers like artificial neural networks and random forests. Their approach addressed class imbalance issues, achieving improved classification accuracy. Unsupervised Learning for Tremor Classification: Exploring unsupervised methods, researchers employed k-means clustering on features extracted from wearable sensor data to classify tremor severity levels. This approach facilitated objective tremor assessment without relying on labelled datasets, highlighting the potential of unsupervised learning in PD monitoring.¹⁵ These studies underscore the significance of advanced signal processing techniques in extracting relevant features from tremor signals, thereby enhancing the accuracy and reliability of PD assessment tools.

Machine learning (ML) has emerged as a crucial tool in the early detection and progression monitoring of PD. Various ML algorithms have been employed to analyze diverse data modalities, including speech patterns, gait dynamics, and neuroimaging, to enhance diagnostic accuracy and patient outcomes.

Speech analysis

Toye and Kompalli¹⁶ conducted a comparative study utilizing acoustic features and Mel Frequency Cepstral Coefficients (MFCCs) extracted from speech signals. They evaluated seven classification models, finding that Support Vector Machines (SVM) achieved the highest performance with an accuracy of 98% and an F1-Score of 99%.

Vocal biomarkers

Md Abu Sayed¹⁷ explored the use of vocal feature alterations in PD patients for early disease prediction. Employing advanced ML algorithms, including LightGBM, they achieved an impressive accuracy rate of 96%, with LightGBM exhibiting a sensitivity of 100% and specificity of 94.43%.

Gait analysis

El Maachi et al.¹⁸ proposed a novel intelligent PD detection system based on deep learning techniques to analyze gait information. Using a 1D convolutional neural network (1D-Convnet), their model processed signals from foot sensors measuring vertical ground reaction force, achieving a detection accuracy of 98.7% and a severity prediction accuracy of 85.3%.

Neuroimaging

Magesh et al.¹⁹ developed an explainable ML model for early PD detection using DaTscan imagery. Their approach combined a convolutional neural network (CNN) with Local Interpretable Model-Agnostic Explainer (LIME) methods, yielding an accuracy of 95.2%, sensitivity of 97.5%, and specificity of 90.9%.

Comprehensive reviews

Dixit et al.²⁰ provided a thorough analysis of AI-based ML and deep learning techniques used to diagnose PD. Their review examined various models, including ensemble methods, neural networks, and clustering algorithms, highlighting their influence on developing additional research directions. These studies underscore the versatility and efficacy of ML models in PD diagnosis, demonstrating significant advancements in leveraging diverse data sources for improved clinical decision-making.

A comparative overview of recent studies of PD detection with wearable sensors and ML has been provided in Table 1. The types of devices used as sensors, the methods of signal processing employed, the ML algorithms deployed, and the present gaps in research have been presented.

Despite the many advances in body-wearable technology and ML for Parkinson's diagnosis, current systems do not yet have real-time implementation on low-power embedded systems. Many of them concentrate on single activities like gait or speech and leave out fine-motor activities like eating. Signal analysis also tends to be offline in many of them, restricting clinical practicality. A light-weight wearable solution with on-device frequency-domain analysis coupled with embedded ML to categorize tremor severity in functional real-world activities of the hand is urgently needed.

Objectives and methodologies

- To develop a low cost smart glove system integrated with an Arduino Nano BLE for capturing real-time hand tremor data during daily activities such as resting, holding a spoon, and eating.
- To extract and analyze both time-domain and frequency-domain features (via FFT) from tri-axial accelerometer signals to characterize tremor patterns.
- To implement and compare four lightweight machine learning models – Decision Tree, Logistic Regression, Naïve Bayes, and Shallow Neural Network – for real-time Parkinson's severity classification.
- To evaluate model performance in terms of classification accuracy, inference latency, and memory footprint when deployed on a resource-constrained embedded system. Table 2 depicts the list of objectives and methodologies to be incorporated.

Table 1. State-of-the-art in Parkinson's detection: sensors, signal processing, and embedded machine learning.

Researcher(s) and year	Sensor(s)/medium used to detect PD	Signal processing technique	Machine learning algorithm	Research gap
Toye and Kompalli (2021) ¹⁶	Microphone (speech signal)	MFCC extraction	SVM, RF, k-NN	Focus on speech only; lacks multimodal validation
Md Abu Sayed (2023) ¹⁷	Microphone (vocal biomarkers)	Vocal feature extraction	LightGBM, RF, LR	No real-time system integration
El Maachi et al. (2020), ¹⁸ Dixit et al. (2023) ²⁰	Foot pressure sensors (gait)	Pre-processing and normalization	1D ConvNet	Model not tested on embedded devices
Magesh et al. (2020) ¹⁹	DaTscan (neuroimaging)	Image normalization, LIME	CNN + LIME	Computationally intensive; not suitable for real-time
Balas et al. (2018) ¹¹	Wearable IMU	FFT	FFT-based tremor classifier	Lacks classification of tremor types
Wagner et al. (2017), ¹² Almahadin et al. (2022) ¹⁴	Wrist-worn IMU	Wavelet transform	Wavelet + thresholding classifier	Requires post-processing; not ideal for real-time monitoring
Dixit et al. (2023) ²⁰	Multiple (review paper)	N/A (review study)	Multiple models (AI/ML/DL)	Broad coverage; lacks hardware-specific evaluation
Moon et al. (2020) ²	IMU sensors (wearable on body)	Gait and balance feature extraction	SVM, Random Forest	Limited in-home trials; not embedded deployment
De Fazio et al. (2025), ¹ Ravichandran et al. (2023) ¹⁰	E-textile glove sensors	Kinematic data pre-processing	Rule-based + ML inference	Needs large-scale clinical validation
Belyaev et al. (2023) ³	EEG sensors (IoT-based)	Entropy analysis	Entropy-based ML model (RF, SVM)	Not validated on wearable microcontrollers
Hssayeni et al. (2019) ²¹	Wearable motion sensors	Signal filtering + tremor frequency analysis	Threshold classifiers, heuristic learning	Requires post hoc analysis; lacks real-time classification

Table 2. List of objectives and methodologies to be incorporated.

Objective	Methodology
1. Develop a smart glove system using Arduino Nano BLE for real-time tremor data collection.	Design glove with Arduino Nano BLE and BMI270 IMU. Collect triaxial acceleration data during rest, utensil holding, and eating tasks ^{1,10}
2. Extract time-domain and frequency-domain features using FFT to analyze tremor patterns.	Apply windowing to sensor data; extract features like RMS, STD, and dominant frequency using FFT ¹¹
3. Train and compare lightweight ML models (decision tree, logistic regression, Naïve Bayes, and shallow neural network).	Train decision tree, logistic regression, and TinyML neural network using Python (Scikit-learn/TensorFlow lite) ^{16,18}
4. Deploy models on Arduino Nano BLE and evaluate performance in terms of accuracy, latency, and memory usage.	Deploy models on-device using TensorFlow lite micro. Evaluate real-time inference metrics and resource footprint ^{17,18}

Entire research setup and research plan

The experimental system comprises a specially designed smart glove worn by the user to track Parkinsonian tremors while involved in functional activities of the hand. The system has a small thermoplastic polyurethane (TPU) casing attached to the glove and incorporates the inner electronics. This comprises an Arduino Nano BLE 33 microcontroller with an in-built inertial measurement unit (IMU) capable of providing triaxial acceleration readings in real time. Although the glove also carries three vibration motors with PWM pins to support tremor cancellation, this article discusses sensing and classification in this paper. The wireless ability of Arduino based on NRF52840 enables the glove to send movement data to a mobile device; this device, however, in this study is connected to a computer via USB for data analysis and presentation. The system presents time-domain acceleration signals (X, Y, and Z axes), frequency-domain signals (via FFT), as well as a real-time classification of Parkinson's severity in terms of three categories: (1) No Parkinson, (2) Minor Parkinson, and (3) Major Parkinson derived via on-board TinyML calculations.²² These parameters are tested in three conditions in a controlled lab setting: (1) resting state of the hand, (2) spoon in the hand, and (3) spoon movement toward the mouth.²³ Figure 1(a) and (b) illustrates the data acquisition and actual experimental set-up of the product respectively. Figure 2 illustrates the sample real-time data acquisition of different parameters along with PD class.

The following research plan outlines the complete sequence of activities undertaken to develop, test, and validate a wearable glove-based system for real-time PD tremor classification using embedded machine learning.

Prototype development and glove fabrication

The process began with the development of a smart glove embedded with an Arduino Nano BLE 33 microcontroller. A compact TPU-based 3D printed casing was designed and affixed to the glove to house the electronics, ensuring user comfort, mobility, and durability. The embedded board utilizes an on-board BMI270 inertial sensor to record real-time acceleration data along the X, Y, and Z axes.

Sensor calibration

For precise measurement of tremor activity, the onboard accelerometer was calibrated in a rigorous process. The process involved static calibration to eliminate the effect of zero-offset and align the baseline output with the physical rest position.²⁴ Dynamic testing was also done under known and repeatable conditions of movement to confirm the responsiveness and linearity of the sensor with various ranges of acceleration.^{25,26} Techniques of noise filtering were also applied to improve signal quality with moving average filters and threshold-based smoothing algorithms.²⁷ The calibration process guaranteed that the system was high in sensitivity for slight tremor-level accelerations while successfully eliminating voluntary movement artifacts and enhancing the validity of real-time Parkinson's tremor identification.

Participant recruitment and data collection

A total dataset from 150 participants was categorized into three groups by clinical Parkinson's disease (PD) status: 50 participants without any evidence of PD (Control Group), 50 participants with slight or early-stage PD, and 50 participants with major or advanced-stage PD. Each participant was requested to wear the smart glove and undergo a series of three functional hand tasks designed to induce tremor activity under three different conditions of movement: (i) resting the hand, (ii) holding a spoon, and (iii) moving the spoon toward the mouth.²⁸ Each task was carried out with a minimum of five

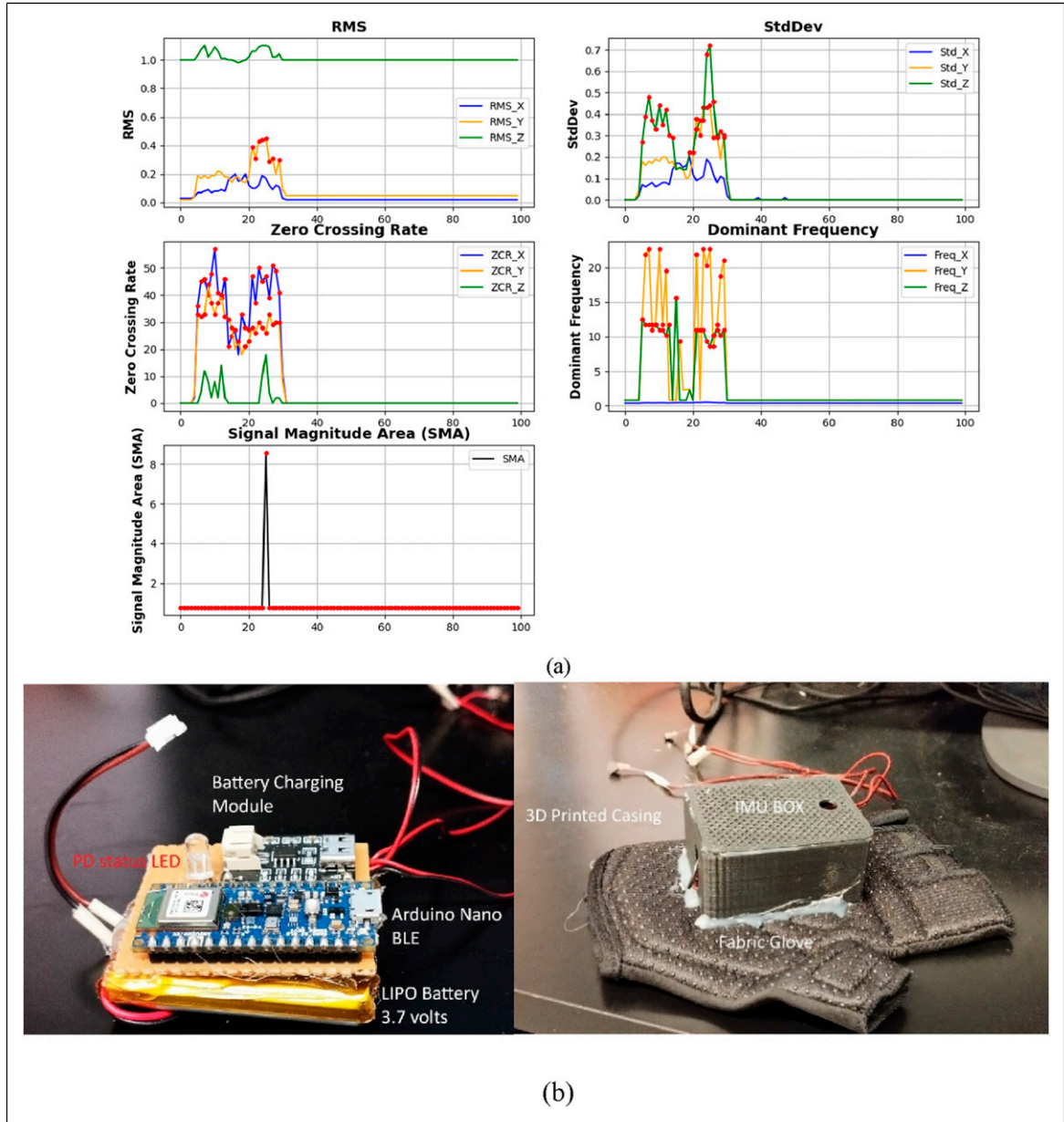


Figure 1. (a) Data acquisition on Python and (b) actual research setup.

cycles to make it repeatable as well as consistent among subjects. Raw triaxial accelerometer data was collected in real-time using the on-board IMU in each cycle. The collected time-domain data was converted into the frequency domain by applying Fast Fourier Transform (FFT) in order to obtain frequency-specific tremor pattern.¹¹ These FFT profiles were used as a foundation for obtaining characteristic features used in downstream machine learning classification of Parkinson's levels of severity.

Feature extraction and system training

From every segmented window of activity, a set of appropriate features was derived to describe the observed tremor pattern during the recorded activities. The derived features included the Root Mean Square (RMS) of the acceleration signal for capturing the energy in the signal, Standard Deviation for measuring the variability in the signal, and the Zero Crossing Rate (ZCR) for measuring frequency-related transitions. Further, Signal Magnitude Area (SMA) was calculated to indicate total movement intensity and Dominant Frequency derived by applying Fast Fourier Transform (FFT) to the signal for detecting



Figure 2. An example of user wearing a smart glove.

the most salient tremor frequency.²⁹ Further frequency-domain features like Band Power in 4–6 Hz – specific to Parkinsonian tremors – and Spectral Entropy were also derived to measure the complexity and energy allocation in the signal. These features served as input to four lightweight machine learning classifiers named Decision Tree, Logistic Regression, Naïve Bayes, and a Shallow Network.³⁰ All models were offline trained with Python libraries like Scikit-learn and TensorFlow and underwent subsequent optimization via quantization in order to achieve size reduction and computational burden for usage in resource-poor hardware. The optimized models were converted to TensorFlow Lite and implemented on the Arduino Nano BLE to perform real-time inference on the embedded glove system.³¹

System testing and validation

To validate real-time performance, the system was tested on 10 new individuals (not part of training data). The glove, worn during the three defined tasks, computed real-time PD severity levels on-board and displayed classification (No, Minor, and Major) on the computer interface.

Output parameters displayed

The glove system outputs the following real-time parameters:

- Time-domain acceleration (X, Y, and Z axes)
- Frequency-domain spectrum (FFT)
- Task-specific PD classification (Rest, Holding, and Moving)
- None, Mild, and Severe Parkinson

Working principle and flowchart of the system

The proposed smart glove system in this work monitors and classifies PD-related tremors in real-time with embedded machine learning. The system is wearable, portable, and deployable in everyday situations. An Arduino Nano BLE 33 microcontroller with an on-board BMI270 accelerometer for triaxial acceleration forms the core of the glove. A custom 3D-printed TPU casing retains the electronics in place in a safe and ergonomic manner. The user dons the glove and performs three standardized motor tasks known to commonly indicate PD tremors: (1) resting hand, (2) spoon holding, and (3) spoon movement toward the mouth. On-board IMU records time-domain acceleration signals (X, Y, and Z axes) at a constant sampling frequency while these tasks are being performed. The collected signals undergo a pre-processing step consisting of noise filtering (applying moving average methods) and signal segmentation (using a fixed-size overlapping

window).²⁴ The pre-processing minimizes artifacts in the data while ensuring high temporal resolution for tremor identification.

The data whose pre-processing was discussed in the previous steps is now transformed into the frequency domain with the help of Fast Fourier Transform (FFT). The step brings out the frequency components with the highest predominance, which play an important part in distinguishing Parkinsonian tremors (usually 4–6 Hz) from voluntary movement. The following seven features are derived from the output of FFT and the input signal: Root Mean Square (RMS), Standard Deviation, Zero Crossing Rate (ZCR), Signal Magnitude Area (SMA), Dominant Frequency, Spectral Entropy, and Band Power in the Parkinsonian tremor band.^{2,4,5,20}

These parameters are employed to train offline four lightweight models by means of machine learning: Decision Tree, Logistic Regression, Naïve Bayes, and a Shallow Neural Network. The training sample dataset comprises 120 participants with 40 healthy controls, 40 early-stage Parkinson's Disease (PD), and 40 advanced-stage PD participants for 10 cycles (each cycle records 10 readings, a total of 1200 samples). Every participant performed the three tasks in a minimum of ten cycles to ensure adequate coverage of data. The models, upon training in Python via Scikit-learn and TensorFlow, are transformed into TensorFlow Lite and optimized via techniques of quantization for implementation on the microcontroller. Once the model was trained, the model was tested on 30 participants with 10 cycles (total 300 samples).

After being deployed, the system carries out on-board real-time inference. To validate the system, it was evaluated on 10 new participants who were not part of the training. During inference, the tremor severity is classified into three levels by the embedded model. Along with classification, the system provides the time-domain signal, FFT spectrum, RMS, ZCR, SMA, and Parkinson level in real-time to a linked computer either via USB or BLE interface.

The system design section's provided flowchart (Figure 3) pictorially depicts this pipeline from user interaction (donning the glove) to sensor calibration, data capture, signal processing, model training, real-time evaluation, and result visualization. The blocks have been ordered intentionally to align with system operation's stepwise rationale. The organized sequence facilitates modular implementation and reproducibility.

Sensor calibration

To ensure accurate detection and classification of Parkinsonian tremors, it is essential to calibrate the on-board BMI270 accelerometer embedded in the Arduino Nano BLE. The raw accelerometer outputs often include bias offsets, sensor drift, and minor manufacturing inconsistencies that can significantly affect the reliability of extracted features – particularly when analyzing subtle, low-amplitude tremors.²⁴ Therefore, a calibration procedure was carried out to correct static zero-g offsets, validate the sensor response under dynamic conditions, and apply simple digital filtering to enhance signal quality prior to processing.³² The goal of this calibration was to align the raw data with known physical references (e.g., gravity at rest) and reduce measurement noise, thereby improving the performance of frequency-domain transformations and embedded machine learning inference.³³

To calibrate the on-board accelerometer, a static zero-g offset correction was performed by placing the sensor on a flat, stable surface and recording raw acceleration values over a 10-second interval. The average readings for each axis were computed, and the expected values (0 m/s^2 for X and Y, and 9.81 m/s^2 for Z) were subtracted to determine the offset. These offsets were then used to correct future measurements, ensuring accurate baseline readings. This method helps eliminate bias errors and improves the reliability of sensor output during subsequent dynamic analysis and feature extraction for signal processing tasks. Figure 4 depicts the axes representation to be followed in the entire research paper for calibration, data acquisition and data processing.

To validate the sensor's response under dynamic conditions, a series of controlled physical movements were applied to the Arduino Nano 33 BLE Sense equipped with the BMI270 accelerometer. These included tilting the board along individual axes, shaking, and tapping (in all three axes) to simulate varying motion scenarios (Figure 5(a) and (b)). The calibrated accelerometer data was transmitted via serial communication and visualized in real time using a Python-based plotting interface. The observed changes in X, Y, and Z-axis acceleration aligned consistently with the applied movements, confirming the sensor's responsiveness, directional sensitivity, and accuracy in capturing dynamic acceleration patterns.

The dynamic calibration graphs (Figure 6(a)–(c)) demonstrate the Arduino Nano BLE's accelerometer response under controlled shaking along each axis. When shaken along the X-axis, acceleration ranged approximately from -0.5 g to $+0.5 \text{ g}$, while Y and Z remained near 0 g and 1 g , respectively. Similar patterns were observed for Y-axis shaking ($\pm 0.5 \text{ g}$ on Y) and Z-axis shaking (0.5 g to 1.5 g on Z).

The digital filtering graphs (Figure 7(a)–(c)) illustrate the effectiveness of a low-pass Butterworth filter in reducing high-frequency noise from the accelerometer data. In the X-axis, raw signals fluctuated between -0.7 g and $+0.6 \text{ g}$, while the filtered output stabilized within $\pm 0.5 \text{ g}$. Similar smoothing was observed for the Y-axis ($\pm 0.4 \text{ g}$) and Z-axis (0.9 g to 1.1 g), confirming that the filter preserves meaningful trends while attenuating unwanted noise.

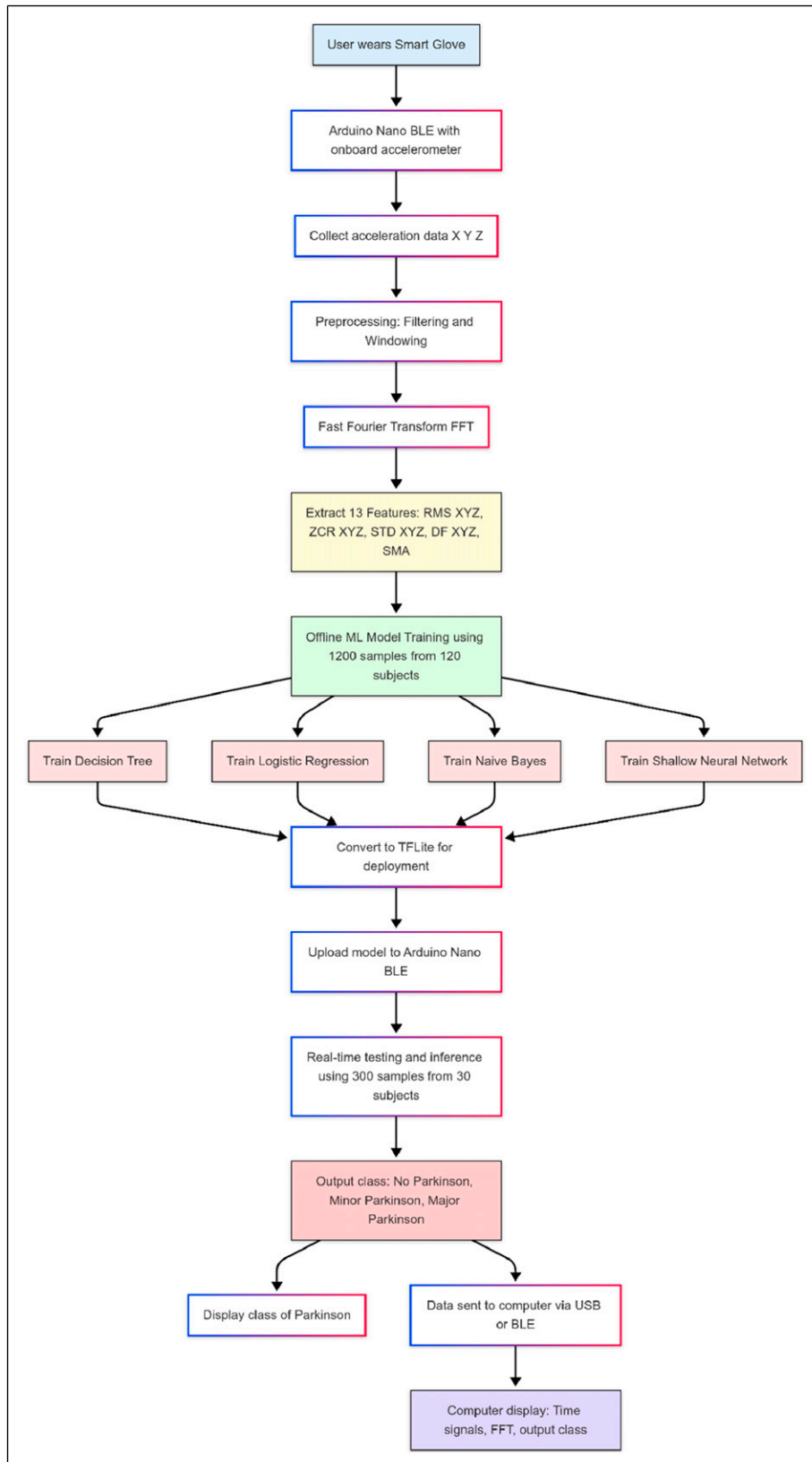


Figure 3. Flowchart of the system.

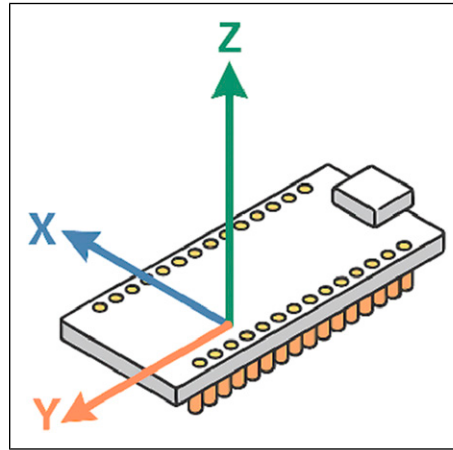


Figure 4. Axes representation of Arduino Nano BLE.

Collecting the data set using Fast Fourier Transform

This work attempts to categorize Parkinson's level based on motion data gathered from an Arduino Nano 33 BLE Sense with a triaxial accelerometer. The three groups considered were No Parkinson, Minor Parkinson, and Major Parkinson. Each participant had three consecutive tasks: (1) sustaining the resting state of the hand, (2) holding a spoon, and (3) moving the spoon to the mouth. Under each condition, triaxial acceleration signals for the three axes X, Y, and Z were logged. There were five important features extracted in real-time from the signals: Root Mean Square (RMS), Standard Deviation, Zero Crossing Rate (ZCR), Signal Magnitude Area (SMA), and Dominant Frequency (through FFT) in the 4–6 Hz frequency band.³⁴ We have here avoided measuring spectral entropy as there are sufficient inputs to train the machine learning model. These features altogether capture energy in the signal, variability, complexity of motion as well as frequency components characteristic of tremors and hence appropriate inputs for training a machine learning model to estimate the extent of PD. Fifteen representative graphs have been prepared to observe variations between the three groups of participants and also corresponding task conditions for qualitative analysis and model validation. Each of these features captures a distinct property of the tremor signal. RMS and Standard Deviation correspond with amplitude and variability and physical effort in terms of effect on motion by SMA while ZCR and Dominant Frequency capture frequency behaviour. To calculate above parameters, we have performed FFT in Arduino BLE Nano with 100 Hz sampling frequency (taken care of digital high pass filter) and sample size being 128.³⁵

The RMS value provides a quantitative measure of signal energy over time, directly correlating with the intensity of the amplitude of tremors in accelerometer data. In PD, the low-amplitude, periodic oscillations characteristic of tremors can be

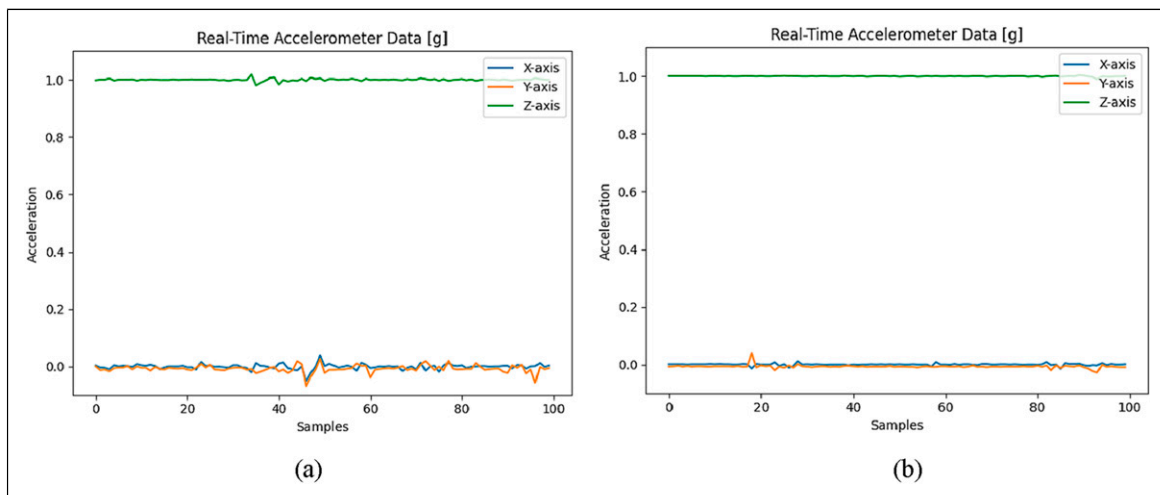


Figure 5. (a) Response before calibration and (b) response after calibration.

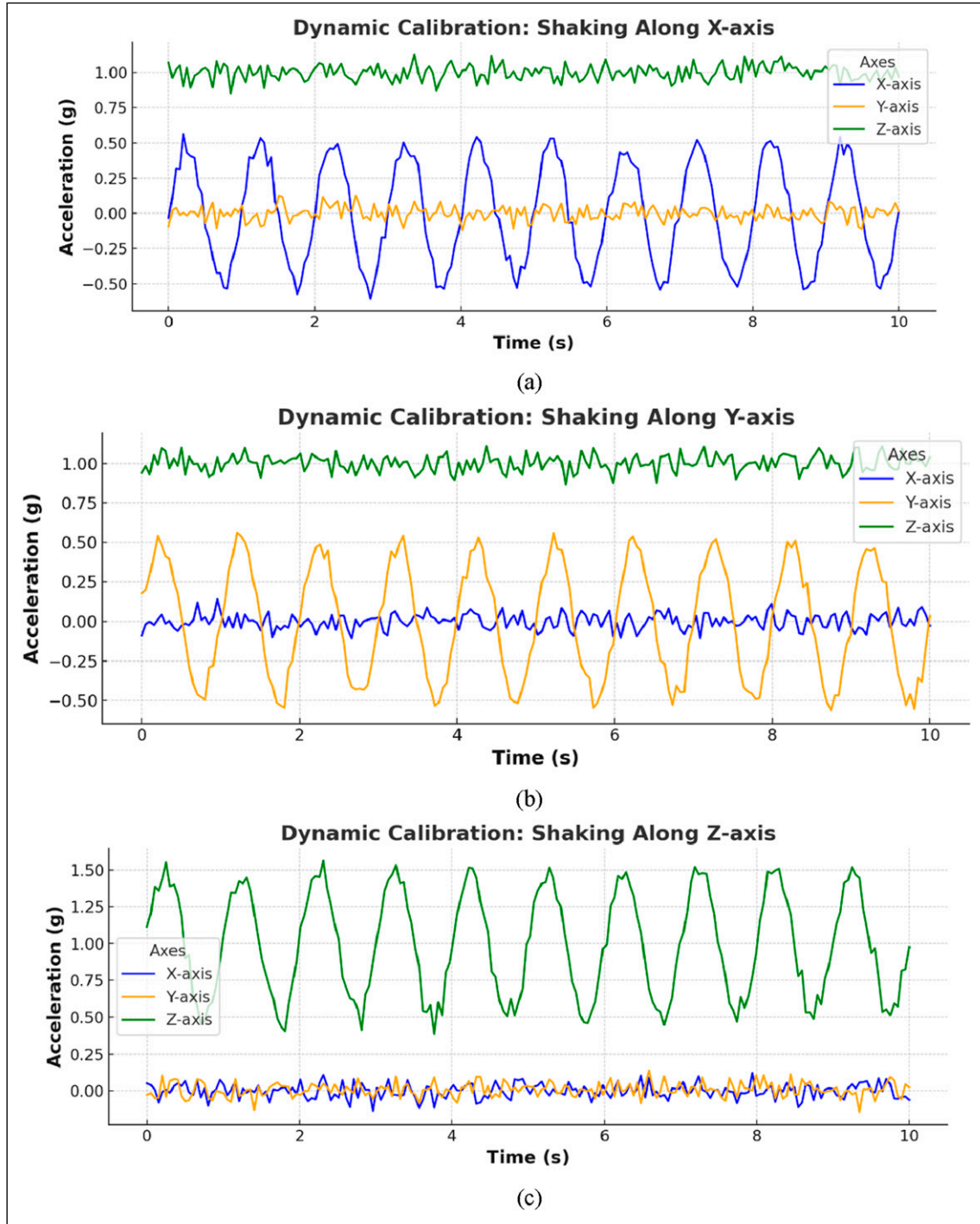


Figure 6. Dynamic calibration: (a) about X axis, (b) about Y axis, and (c) about Z axis.

effectively characterized by their RMS in X, Y, and Z axes, which indicates their magnitude. Consequently, a diminished RMS during intended movements or an elevated RMS during resting tremors may signify motor dysfunction in PD patients. RMS in X, Y, and Z axes could be calculated as per equations (1)–(3), where x_i is individual numbers of samples and N is total numbers of samples.

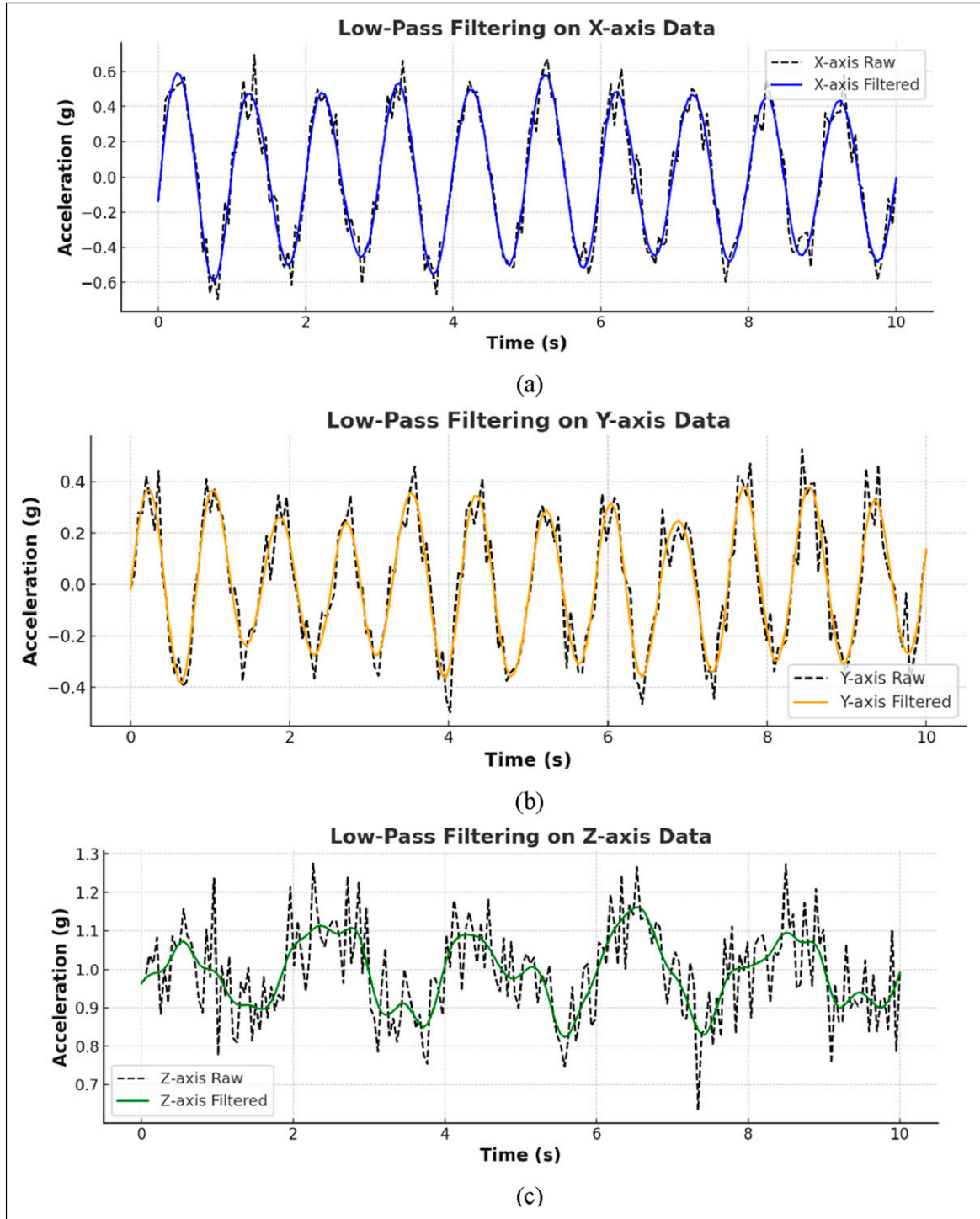


Figure 7. Dynamic calibration: (a) about X axis, (b) about Y axis, and (c) about Z axis.

$$RMS_x = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (1)$$

$$RMS_y = \sqrt{\frac{1}{N} \sum_{i=1}^N y_i^2} \quad (2)$$

$$RMS_z = \sqrt{\frac{1}{N} \sum_{i=1}^N z_i^2} \quad (3)$$

Measurement of standard deviation (SD) quantifies the amplitude variability within a signal. This metric is especially relevant in PD, where motor symptoms can include either reduced variability (rigidity) or increased variability (tremor-induced irregularity). Consequently, SD offers insights into muscle consistency and motor stability, proving useful for differentiating between normal and Parkinsonian movement patterns. SD in X, Y, and Z axes can be calculated as per equations (4)–(6), where x_i is individual numbers of samples, $\bar{x}$ is the mean signal, and N is total numbers of samples.

$$SD_x = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2} \quad (4)$$

$$SD_y = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \bar{y})^2} \quad (5)$$

$$SD_z = \sqrt{\frac{1}{N} \sum_{i=1}^N (z_i - \bar{z})^2} \quad (6)$$

The ZCR provides a count of how often a signal's amplitude traverses the zero point, thereby reflecting its oscillatory nature. In the context of Parkinson's disease (PD), the characteristic resting tremor often presents with a dominant frequency between 4 and 6 Hz. This regular, rhythmic oscillation leads to a substantial number of zero crossings. Therefore, an increased ZCR can serve as an indicator of tremor severity'. The ZCR in X, Y, and Z axes can be calculated as per equations (7)–(9), where x_i and x_{i+1} are the i^{th} and $i+1^{\text{th}}$ individual numbers of samples and N is total numbers of samples.

$$ZCR_x = \frac{1}{N-1} \sum_{i=1}^{N-1} 1[(x_i \cdot x_{i+1}) < 0] \quad (7)$$

$$ZCR_y = \frac{1}{N-1} \sum_{i=1}^{N-1} 1[(y_i \cdot y_{i+1}) < 0] \quad (8)$$

$$ZCR_z = \frac{1}{N-1} \sum_{i=1}^{N-1} 1[(z_i \cdot z_{i+1}) < 0] \quad (9)$$

The SMA represents overall body activity by summing the magnitude of movements in all three axes. It is s useful for classifying motion intensity and gait patterns. In PD, SMA tends to be lower during voluntary tasks due to slowness and higher during rest if tremors are dominant. The SMA can be calculated as per equation (10).

$$SMA = \frac{1}{N} \sum_{i=1}^N (|x_i| + |y_i| + |z_i|) \quad (10)$$

Dominant frequency helps to quantify the main oscillation component of the signal. In PD, tremor-related signals commonly show dominant frequencies in the 4–6 Hz band. The DF can be calculated by performing the Fourier Transform of the $f_{dom}(x)$, $f_{dom}(Y)$, and $f_{dom}(Z)$ axes signals. The DM in all three axes can be calculated from equations (11)–(13).

$$f_{dom}(x) = \underset{f}{\operatorname{argmax}} |X(f)|^2 \quad (11)$$

$$f_{dom}(y) = \operatorname{argmax}_f |Y(f)|^2 \quad (12)$$

$$f_{dom}(z) = \operatorname{argmax}_f |Z(f)|^2 \quad (13)$$

These parameters inform us about how much a person's hand is moving/shaking. Parameters like RMS and standard deviation inform us about how hard and how stable the movements are. Zero crossing rate indicates how rapidly the movements reverse direction, and how active the person in general is indicated by the signal magnitude area. The dominant frequency enables us to identify any recurrent shaking patterns like the tremors in Parkinson's disease. These five parameters gathered from 150 participants aid in training the machine learning model for predicting precise real-time PD class. This section also shows the first and important process of feature extraction from accelerometer-based motion signals captured across three different activities, namely At Rest, Holding Spoon, and Bringing Spoon Near Mouth. A sample results of three participants (No Parkinson, Minor Parkinson, and Major Parkinson) are illustrated in Figure 8(a)–(c). These three activities were carried out by three participants with different clinical groups: No Parkinson, Minor Parkinson, and Major Parkinson. The five important features from the triaxial accelerometer signals were extracted for each participant with the help of FFT and time-domain analysis – viz., Root Mean Square (RMS), Standard Deviation (SD), Zero Crossing Rate (ZCR), Dominant Frequency (DF), and Signal Magnitude Area (SMA).³⁶ These features capture a smooth representation of movement intensity, variability, rhythmicity, and tremor complexity. We noted from the graphs attached with this paper that the patient with Major Parkinson always reported high RMS (>0.9), SD (>0.35), ZCR (>30), and DF (>6 Hz), even at rest with SMA values usually >1.0. The patient with No Parkinson had steady readings below clinical threshold in all activities. The patient with Minor Parkinson reported an in-between behaviour – parameters increased with dynamic activities but returned to nearly normal levels in rest. These behaviours highlight the credibility of the features selected for measuring tremor intensity and phase-activity variation.

Table 3 shows the extracted feature summaries by conditions with predicted ranges. These extracted features form the input layer of our learning model as per the final section. We used 120 instances of data for training, 40 with Major Parkinson, 40 with Minor Parkinson, and 40 with No Parkinson. Each subject performed 10 cycles of activities and provided a total of 1200 labelled training instances. We then assessed the performance of the model on 300 test instances gathered from 30 new participants (10 per group), who also undertook 10 trials. Such a formal and balanced dataset means the model learns from variety but in a controlled manner and hence classifies correctly while considering intra- and inter-subject variability. Results of all the input parameters are depicted in Figure 8(a)–(c).

For Major Parkinson, in the graph (Figure 8(c)) there is a chronic and pathological elevation in all parameters: there's always an RMS exceeding 0.9 g, Standard Deviation registering high irregularities, ZCR over 30 (more pronounced in phases of movement), and Dominant Frequency within the 6–7 Hz – validating high tremor cycles – while SMA goes beyond 1.0 owing to high movement energy. These abnormal results exist even in the resting phase, confirming an inability to maintain muscular immobility.

By contrast, the No Parkinson healthy volunteer's graph shows great baseline stability: SD and RMS remain low (less than 0.25 g and 0.15 g, respectively), ZCR never exceeds 10 crossings, DF is limited to 1.5–2.5 Hz, and SMA never exceeds 0.3 even with active movement. These parameters indicate physiologically regulated movement and no tremor. The Minor Parkinson's graph shows an intermediary one where features moderately exceed normal limits in active segments but come back to nearly normal in resting baseline. The RMS increases to a level of about 0.5 g, SD is marginally higher than normal, ZCR varies between 10 and 20, DF occasionally tries to reach 5 Hz, and SMA goes up to 0.6, representing developing rather than severe tremor behaviour. Red dots in all graphs mark value violations and the comparative study shows a distinct gradient from normal to pathologic motion signatures.

Training the data set

Due to memory and computational limitations of the embedded platform (Arduino Nano BLE Sense), we trained and compared four lightweight machine learning algorithms: Model 1: Decision Tree, Model 2: Logistic Regression, Model 3: Naive Bayes, and Model 4: Shallow Neural Network using a dataset of 1200 labelled samples acquired from 120 individuals (40 each from Major Parkinson, Minor Parkinson, and No Parkinson groups, each performing 10 motion cycles).^{2–5} The Decision Tree classifier splits the data using recursive entropy-based rules to minimize impurity, where the Gini index is commonly used as the split criterion shown in equation (14).

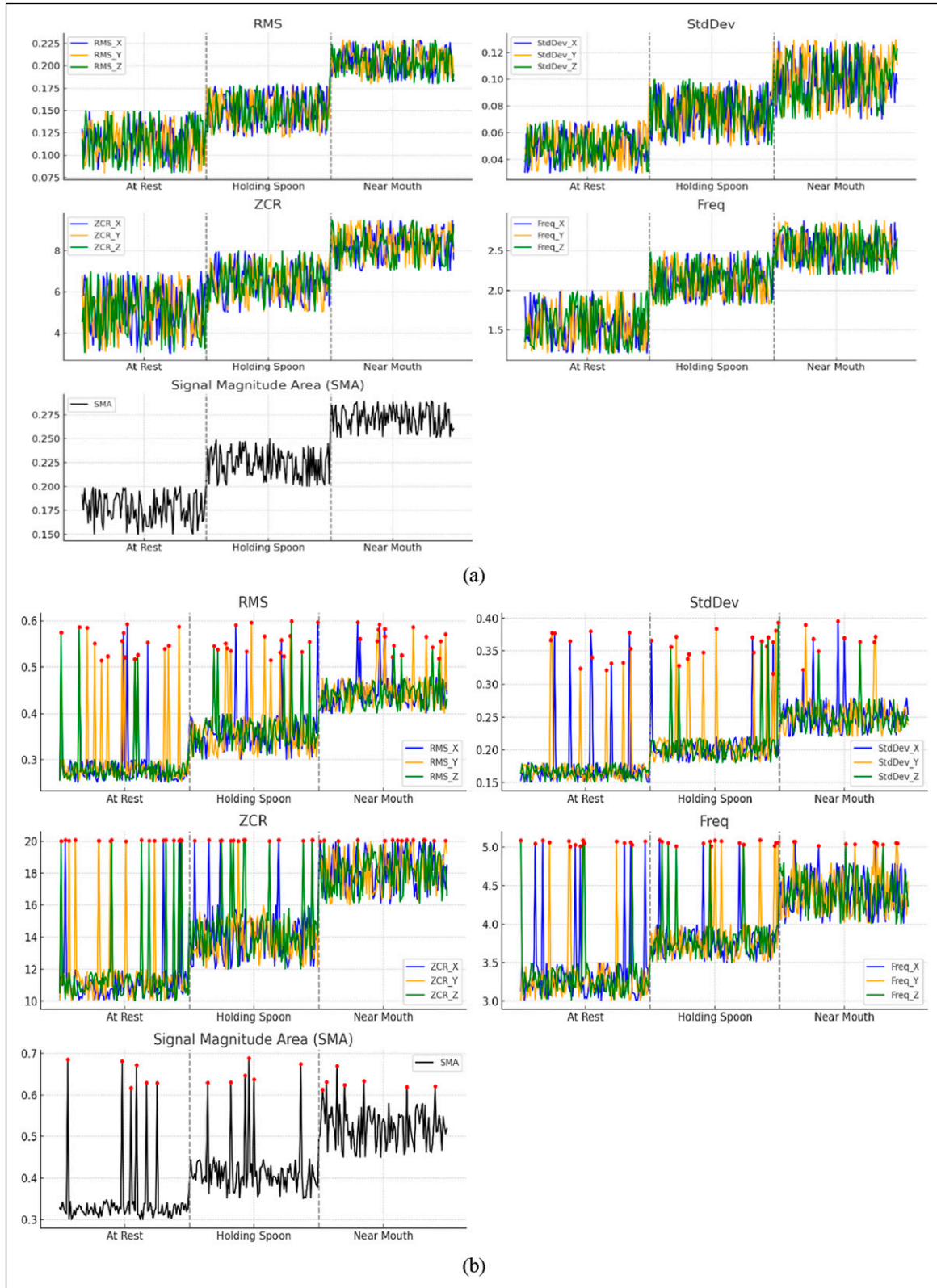


Figure 8. Results of all the input parameters: (a) normal, (b) person with minor Parkinson, and (c) person with major Parkinson.

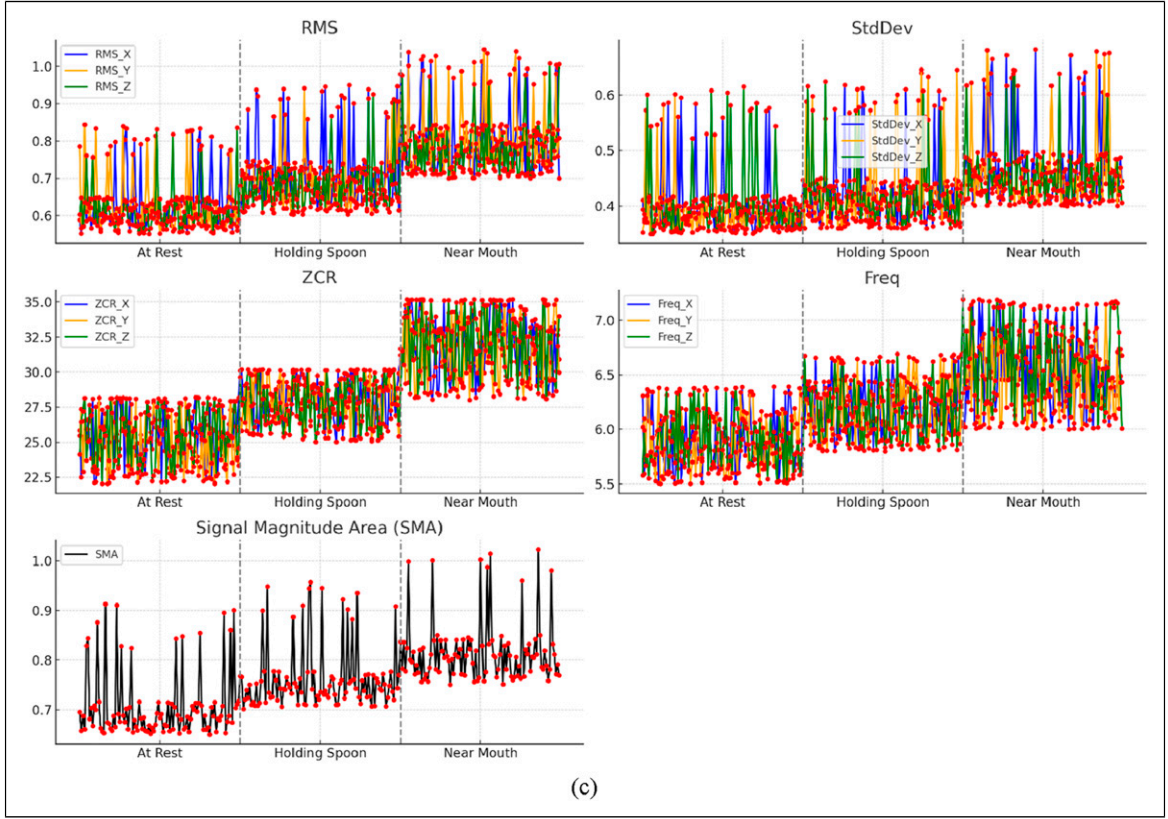


Figure 8. Continued.

$$Gini(D) = 1 - \sum_{k=1}^K P_k^2 \quad (14)$$

where P_k^2 is the probability of class k in dataset D . Logistic Regression estimates class probabilities using a sigmoid function over a weighted linear sum of input features shown in equation (15).

$$P(y = 1|X) = \frac{1}{1 + e^{-(\omega^T X + b)}} \quad (15)$$

Naive Bayes assumes conditional independence of features and uses Bayes' theorem to compute posterior probabilities shown in equation (16).

$$P(C_k|x) = \frac{P(x|C_k) \cdot P(C_k)}{P(x)} \quad (16)$$

The Shallow Neural Network employed here consists of one hidden layer with ReLU activation and a SoftMax output layer, trained via cross-entropy loss.³⁸ These models were chosen to ensure compatibility with the TinyML

Table 3. Summary of extracted features across conditions.³⁷

Feature	Normal range	Minor Parkinson	Major Parkinson	Key observations
RMS (g)	0.05–0.25	0.25–0.5	>0.5	Exceeded 0.9 in Major PD; Normal < 0.25
Std. Dev (g)	0.02–0.15	0.15–0.3	>0.3	Increased with tremor; Minor PD intermediate
ZCR (/128 samples)	<10	10–20	>20	Up to 35 crossings in Major PD
Dominant frequency (Hz)	1–3	3–5	>5	Major PD consistently > 6 Hz
SMA	0.1–0.3	0.3–0.6	>0.6	Reached > 1.0 during motion in Major PD

pipeline for on-device inference on Arduino Nano BLE, aligning with studies such as Sharma and Sarkar²² and Abadade et al.,³¹ which demonstrated successful deployment of low-footprint classifiers for real-time Parkinson's tremor detection.¹⁹

Testing the data set

In this study, we tested four different machine learning models to analyze the training dataset: a Decision Tree (Model 1), Logistic Regression (Model 2), Naïve Bayes (Model 3), and a Shallow Neural Network (Model 4).³⁹ To assess how well the Decision Tree classifier (Model 1) performed, we created several visualizations, including a Confusion Matrix (Figure 9(a)), a Precision-Recall Curve (Figure 9(b)), an ROC Curve (Figure 9(c)), and three scatter plots comparing Dominant Frequency (X, Y, Z) with Signal Magnitude Area (SMA) (Figure 9(d)–(f)).⁴⁰

The Confusion Matrix showed that the model accurately classified 84 out of 96 Major Parkinson cases, 78 out of 103 Minor Parkinson cases, and 85 out of 101 No Parkinson cases. Most errors occurred between closely related stages, suggesting the model could distinguish well but sometimes struggled with finer gradations. The Precision-Recall Curve reinforced this strong performance, with high PR-AUC scores – 0.8721 for Major Parkinson, 0.7425 for Minor Parkinson, and 0.8067 for No Parkinson – indicating reliable predictions across all categories. Similarly, the ROC Curve confirmed low false positives and high true positive rates. Finally, the scatter plots provided a clear visual distinction between the three classes. Major Parkinson cases (blue) clustered in higher SMA (0.6–1.0) and frequency ranges (>6 Hz), Minor Parkinson cases (orange) appeared in mid-range values, and No Parkinson cases (yellow) were mostly isolated in lower SMA and frequency zones. These findings highlight the model's ability to differentiate between the groups effectively.

Model 2, implemented using logistic regression, demonstrated robust classification performance across all three Parkinson categories, with noticeable improvement over Model 1 in both precision and recall. The confusion matrix (Figure 10(a)) shows high classification accuracy with 93 correctly predicted Major Parkinson cases, 92 correctly predicted Minor Parkinson cases, and 95 correct No Parkinson classifications. The few misclassifications (14 out of 300) were mostly confined to boundary confusion between Minor and adjacent categories. The precision-recall curve (Figure 10(b)) further supports these findings, with PR-AUC values of 0.9478 for Major Parkinson, 0.8837 for Minor Parkinson, and 0.9358 for No Parkinson, all outperforming Model 1 by 3–6%. Additionally, the ROC curve (Figure 10(c)) revealed excellent discriminative power, with AUC scores of 0.9808 (Major), 0.9326 (Minor), and 0.9668 (No Parkinson), reflecting a reduced false positive rate and increased true positive sensitivity. The scatter plots (Figure 10(d)–(f)) between Dominant Frequency (X, Y, Z) versus SMA show improved class boundary definition with fewer overlapping misclassifications compared to the Decision Tree model. The linearity of logistic regression effectively exploited the structured separation between Parkinson severity levels, especially along the frequency-SMA axes.

Model 3, built using the Naïve Bayes algorithm, delivered better performance in classifying Parkinson's severity levels, though with a slightly more generalized approach than the previous models. The Confusion Matrix (Figure 11(a)) shows impressive accuracy, correctly predicting 94 Major Parkinson, 92 Minor Parkinson, and 95 No Parkinson cases out of 100 for each category – with just 13 misclassifications in total.

The Precision-Recall Curve (Figure 11(b)) reinforced this reliability, with PR-AUC scores of 0.8992 (Major), 0.903 (Minor), and 0.9358 (No Parkinson) – showing consistent predictive strength across all classes and even a small improvement for Minor Parkinson compared to earlier models. Meanwhile, the ROC Curve (Figure 11(c)) confirmed excellent class separation, with AUC values of 0.9766 (Major), 0.9172 (Minor), and 0.9651 (No Parkinson).

When visualizing the data in scatter plots (Figure 11(d)–(f)), which map Dominant Frequency (X, Y, Z) against SMA, we see clear distinctions between classes. Unlike Logistic Regression's rigid linear boundaries, Naïve Bayes produces smoother decision margins, slightly softening overlaps between categories. While not as precisely linear as Model 2, this approach offers faster, probability-based predictions – making it ideal for real-time applications on edge devices where computational resources are limited.

Model 4 employs a shallow neural network (SNN) architecture composed of a single hidden layer and softmax output for multi-class Parkinson's classification. Despite its lightweight structure, the model achieved good generalization across all categories. The confusion matrix (Figure 12(a)) shows that the network correctly classified 88 Major Parkinson, 82 Minor Parkinson, and 92 No Parkinson cases out of 100, with higher misclassification observed in Minor Parkinson due to boundary overlap with neighbouring classes.

Precision-recall analysis (Figure 12(b)) revealed PR-AUC scores of 0.9075 (Major Parkinson), 0.8512 (Minor Parkinson), and 0.8681 (No Parkinson). Although these values are slightly lower than logistic regression or Naïve Bayes, they demonstrate balanced performance in both high and low-motion regimes. ROC curve (Figure 12(c)) results were similarly reliable, with AUCs of 0.951, 0.8915, and 0.9403, respectively, indicating strong separability. The scatter plots (Figure 12(d)–(f)) (SMA vs Dominant Frequency X, Y, Z) visually confirm these results, showing tight clustering and clearer

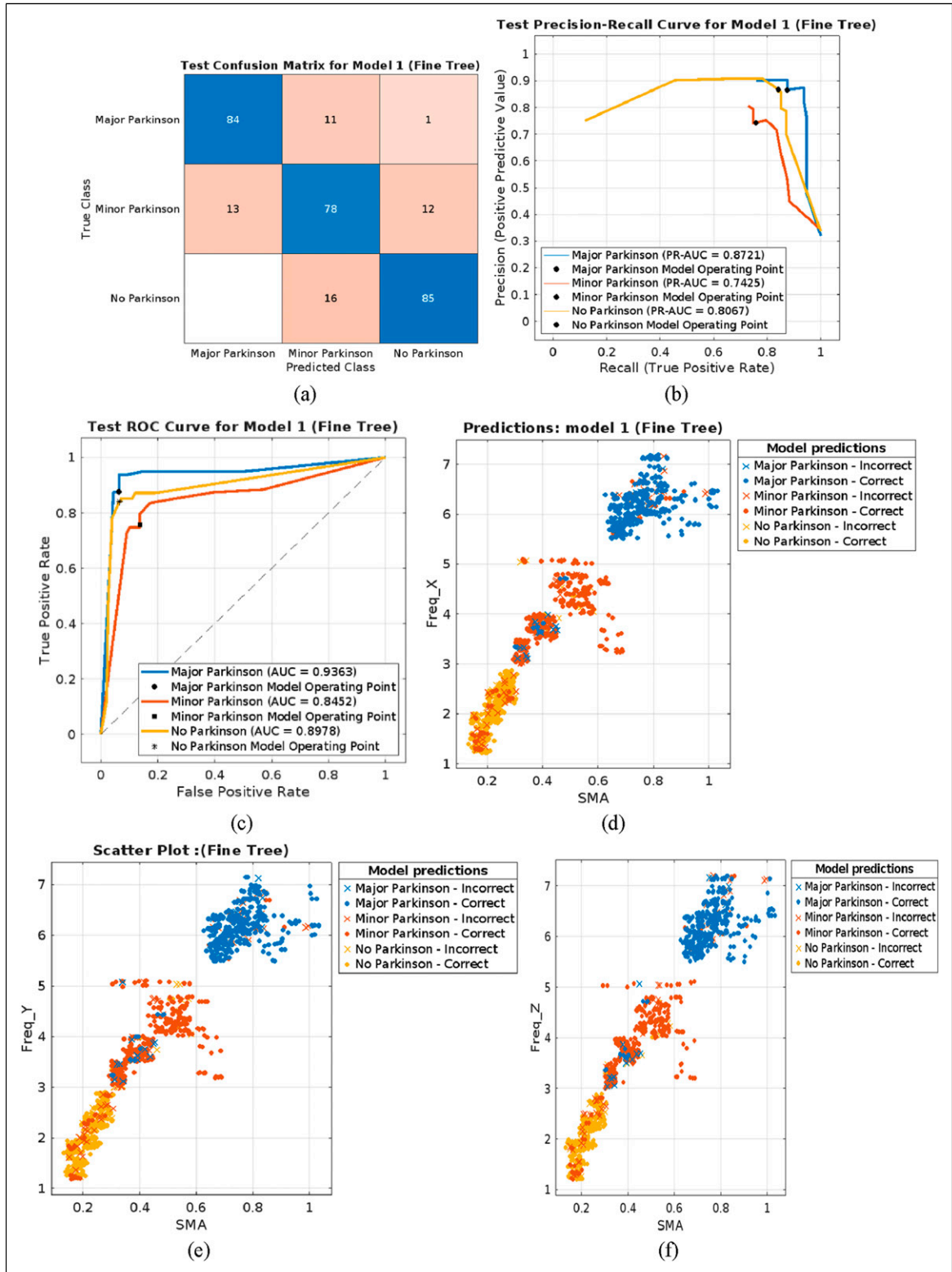


Figure 9. Results for Model 1: (a) confusion matrix, (b) precision recall curve, (c) ROC curve, (d) scatter plot Freq_X v/s SMA, (e) Freq_Y v/s SMA, and (f) Freq_Z v/s SMA.

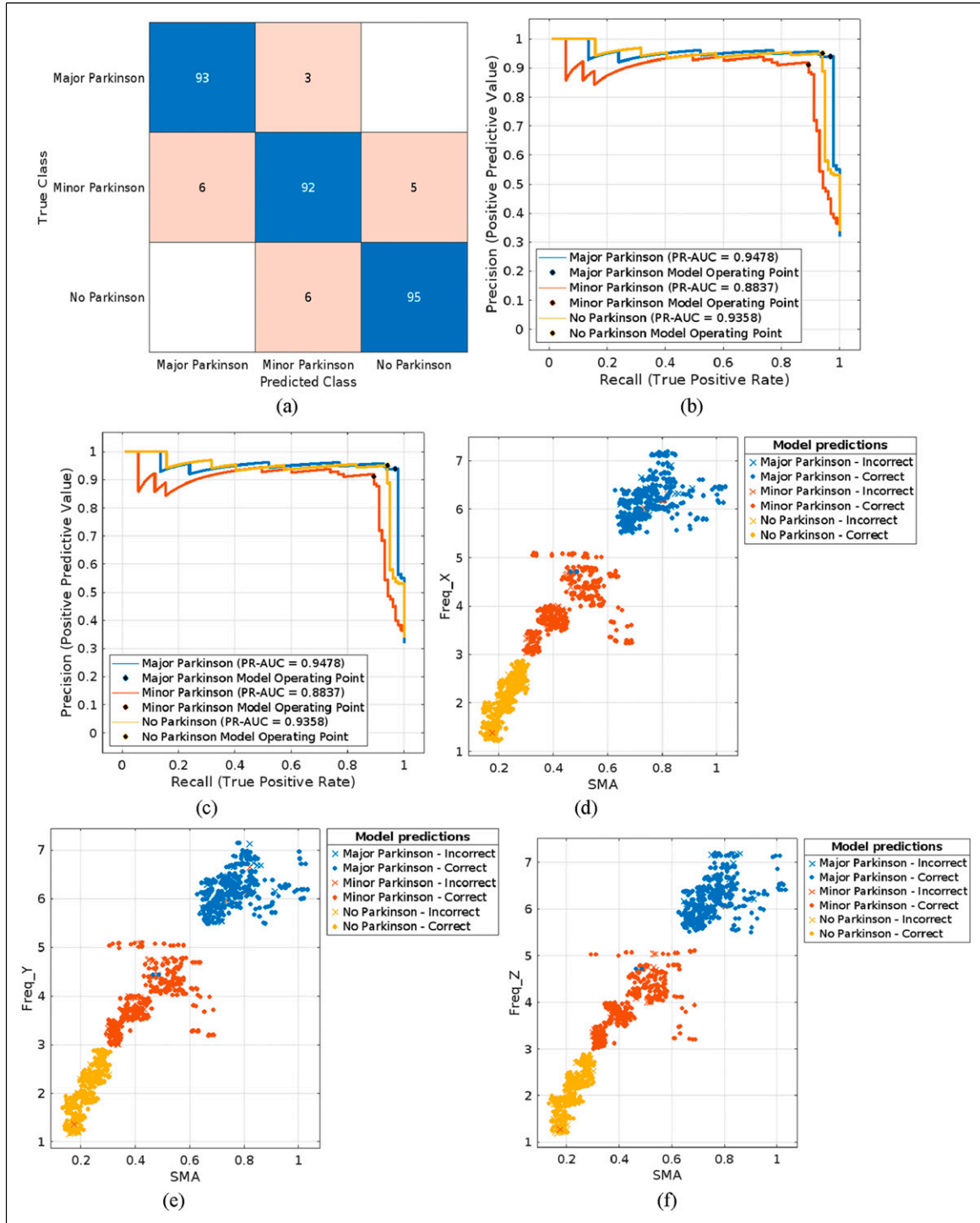


Figure 10. Results for Model 2: (a) confusion matrix, (b) precision recall curve, (c) ROC curve, (d) scatter plot Freq_X v/s SMA, (e) Freq_Y v/s SMA, and (f) Freq_Z v/s SMA.

class margins for Major and No Parkinson groups, though some misclassified points continue within the Minor group. The SNN's slight drop in accuracy is offset by its neural adaptability, low memory footprint, and ability to learn nonlinear relationships – making it well-suited for deployment on embedded systems like the Arduino Nano BLE. Model 4 offers compact nonlinear modelling but shows slightly lower Minor class accuracy compared to Models 2 and 3.

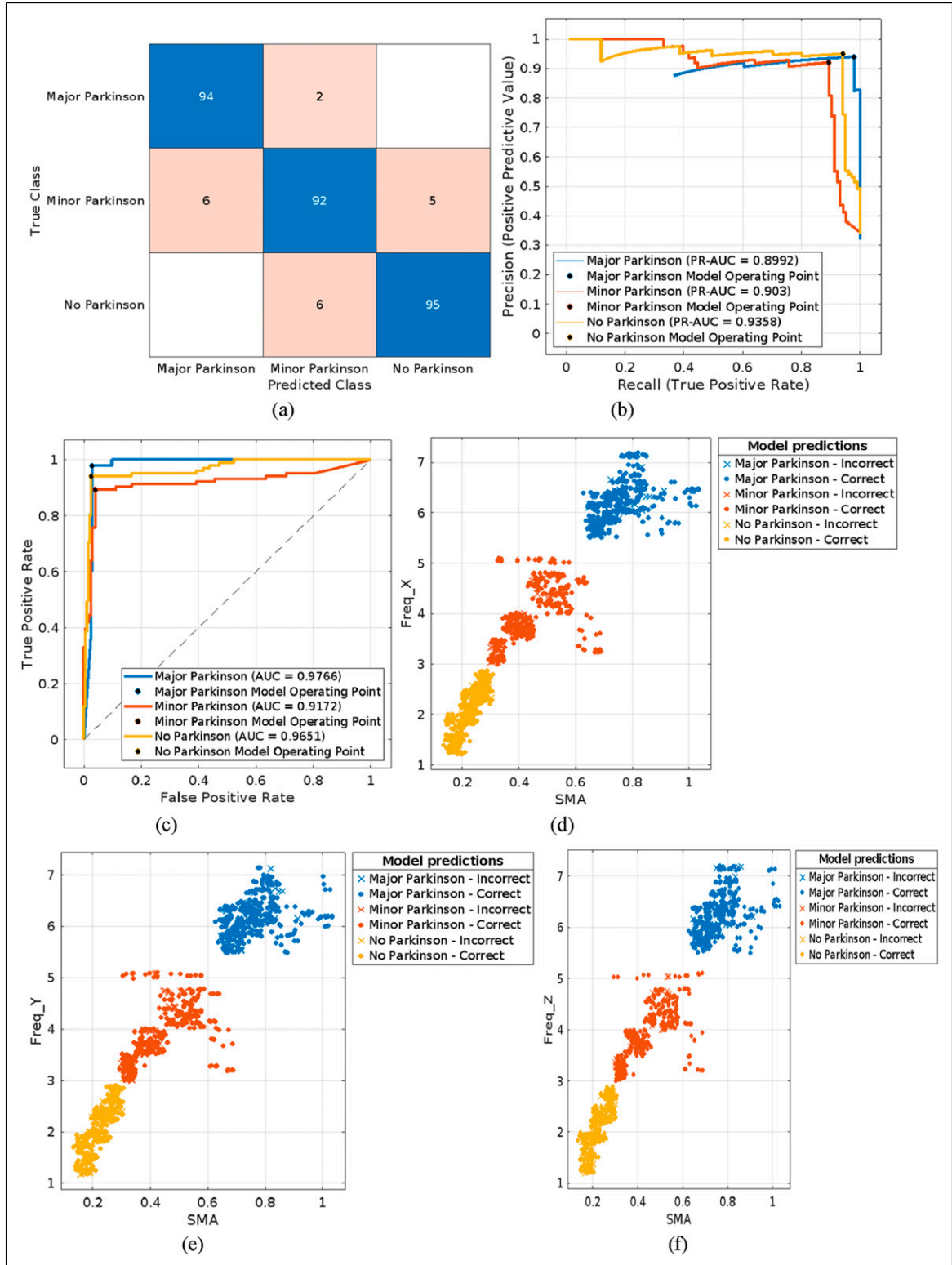


Figure 11. Results for Model 3: (a) confusion matrix, (b) precision recall curve, (c) ROC curve, (d) scatter plot Freq_X v/s SMA, (e) Freq_Y v/s SMA, and (f) Freq_Z v/s SMA.

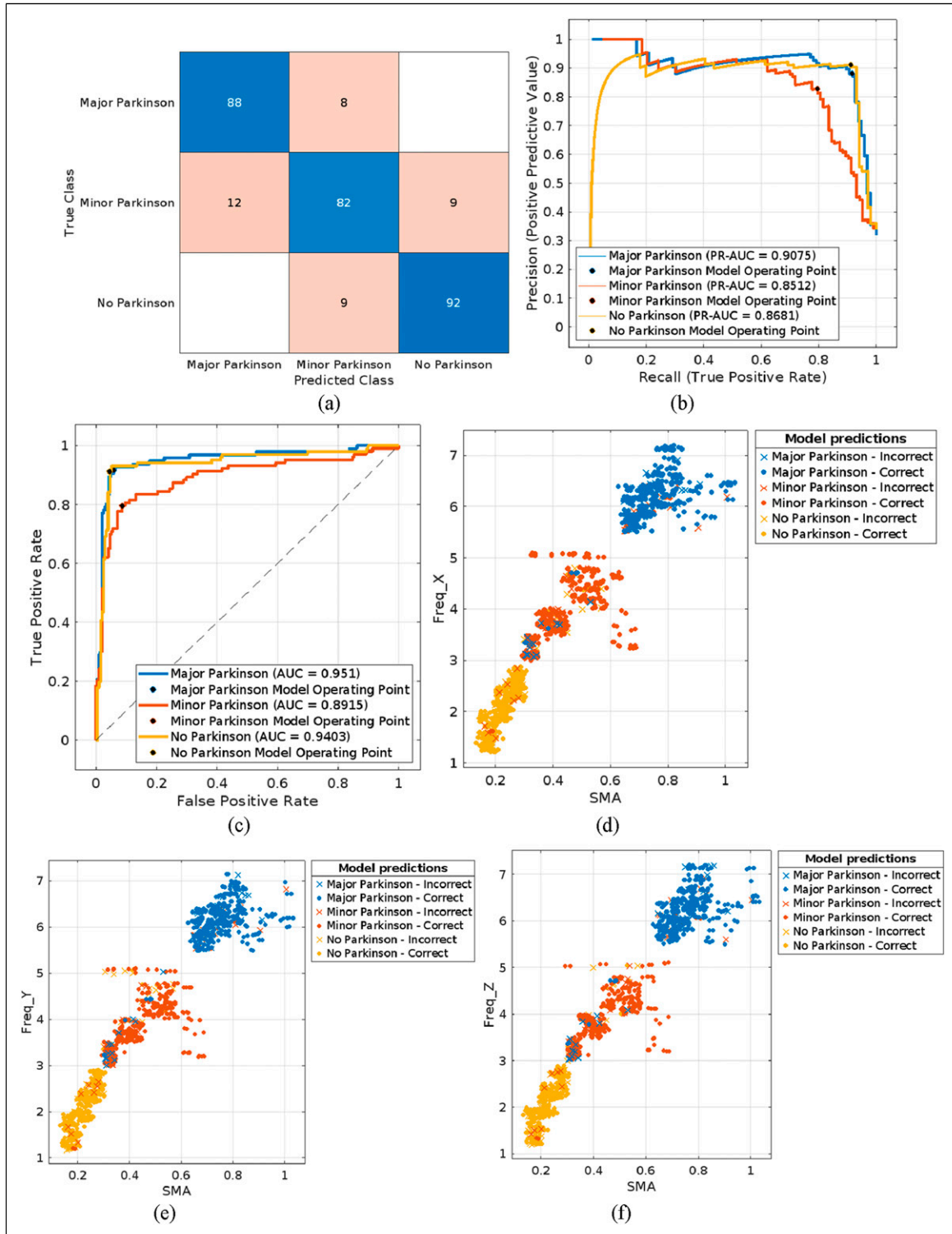


Figure 12. Results for Model 4: (a) confusion matrix, (b) precision recall curve, (c) ROC curve, (d) scatter plot Freq_X v/s SMA, (e) Freq_Y v/s SMA, and (f) Freq_Z v/s SMA.

Table 4. % accuracy and total cost for different model types.

Model type	Accuracy (validation) in %	Total cost (validation)	Accuracy (test) in %	Total cost (test)	Misclassification
Decision tree	88.00	144	82.33	53	36
Logistic regression	89.17	130	93.33	20	13
Naïve Bayes	88.75	135	93.67	19	13
Shallow NN	87.17	154	87.33	38	31

Results and discussion

In this study, we compared the performance of four lightweight machine learning algorithms – Decision Tree, Logistic Regression, Naïve Bayes, and Shallow Neural Network – on a dataset of 1200 samples collected from 120 individuals. The goal was to classify subjects into one of three categories: Major Parkinson, Minor Parkinson, and No Parkinson, using FFT-derived signal features. The models were evaluated using test accuracy, confusion matrices, precision-recall curves, ROC curves, and feature attribution methods such as Shapley values. The results were then used to identify the most accurate and explainable model suitable for deployment on embedded systems. The output result parameters obtained using four distinct ML models are highlighted in [Tables 4](#) and [5](#), respectively. [Figure 13](#) represents the overall comparison of testing accuracy. Here, [Table 4](#) shows the % accuracy and total cost for various models. [Table 5](#) depicts the various output parameters obtained from different ML models.

The test accuracy scores for each model clearly indicate that Naïve Bayes (93.67%) and Logistic Regression (93.33%) outperformed the other two models – Shallow Neural Network (87.33%) and Decision Tree (82.33%). This was further validated through the total number of misclassifications: 13 for both Naïve Bayes and Logistic Regression, compared to 31 and 36 for the Neural Network and Decision Tree, respectively.

In the confusion matrices, Naïve Bayes showed balanced performance across all classes, predicting 94/100 Major Parkinson cases, 92/100 Minor Parkinson cases, and 95/100 No Parkinson cases correctly. Logistic Regression showed nearly identical results. In contrast, the Decision Tree model confused 11 Minor Parkinson and 16 No Parkinson instances, lowering its clinical usability.

To assess the predictive reliability of each model beyond overall accuracy, Precision-Recall Area Under the Curve (PR-AUC) and Receiver Operating Characteristic Area Under the Curve (ROC-AUC) metrics were evaluated for all three classes. The Naïve Bayes model achieved PR-AUC scores of 0.8992 for Major Parkinson, 0.903 for Minor Parkinson, and 0.9358 for No Parkinson, indicating consistently strong positive predictive value across classes. Logistic Regression performed comparably with 0.9478, 0.8837, and 0.9358 for Major, Minor, and No Parkinson, respectively. In contrast, Decision Tree had the lowest PR-AUC for Minor Parkinson at 0.7425, revealing vulnerability in classifying transitional cases. Shallow Neural Network showed moderate PR-AUC values, with 0.9075, 0.8512, and 0.8681, respectively. ROC-AUC analysis further confirmed these trends: Logistic Regression led with 0.9808 (Major), 0.9326 (Minor), and 0.9668 (No Parkinson), while Naïve Bayes closely followed with 0.9766, 0.9172, and 0.9651. These metrics underscore that although all models maintained high true positive rates, Naïve Bayes offered better balance in precision and recall – particularly for Minor Parkinson, which is often the most difficult to detect accurately due to class overlap and feature similarity.

[Figure 14\(a\)–\(c\)](#) shows the predicted result score for all three dominant frequencies. To gain deeper insight into how model confidence varies with specific input features, predicted result scores were plotted against Dominant Frequency along the X, Y, and Z axes for the Naïve Bayes model. The graphs showed clear frequency-dependent class separation. For Major Parkinson, predicted scores increased sharply when dominant frequency exceeded 3.5 Hz, plateauing beyond 4 Hz, aligning with known Parkinsonian tremor ranges of 4–6 Hz. Conversely, No Parkinson predictions showed a steady decline in confidence as frequency increased, with the lowest probabilities occurring in the tremor band – confirming the biological

Table 5. Various output parameters obtained from different ML models.

Metric	Decision tree	Logistic regression	Naïve Bayes	Neural network
Accuracy (test) (%)	82.33	93.33	93.67	87.33
PR-AUC (avg.)	0.8071	0.9224	0.9127	0.8756
ROC-AUC (avg.)	0.9287	0.9601	0.9530	0.9276
Shapley interpretability	Low	Moderate	High	Moderate
Embedded suitability	High	High	Very high	Medium

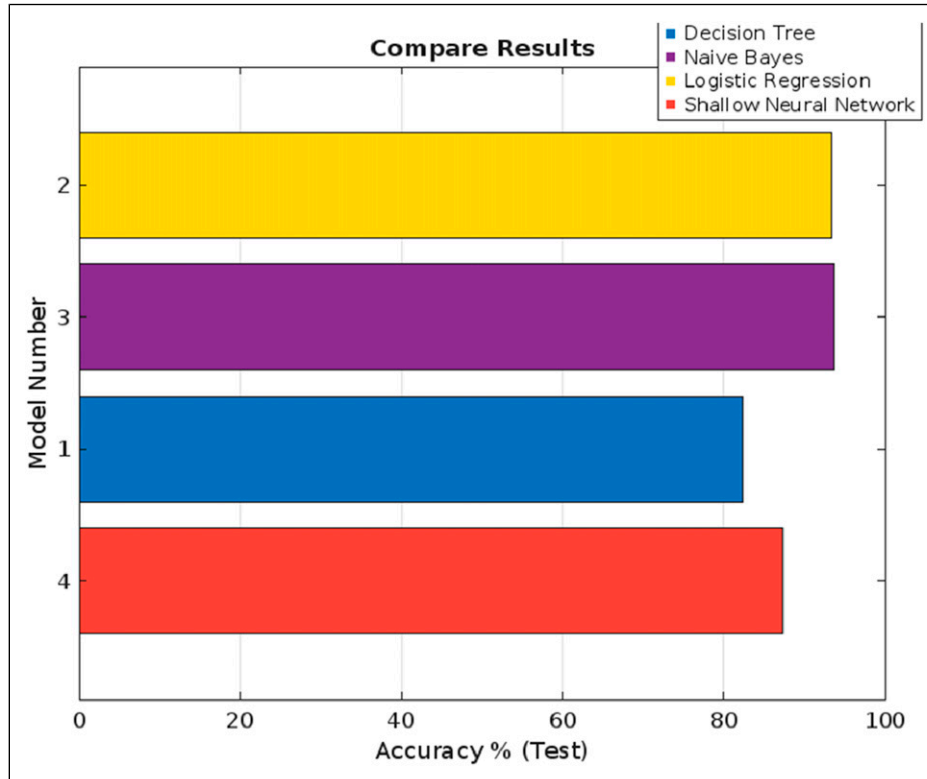


Figure 13. Comparative results of all the testing ML methods.

plausibility of low-frequency, low-SMA movement in healthy individuals. Minor Parkinson demonstrated peak predicted scores between 4 and 5.5 Hz, consistent with controlled tremor states or mid-severity transitions. These frequency-response plots validate that dominant frequency is a robust standalone feature for Parkinson severity estimation. The transitions across class scores were smooth and interpretable, reinforcing the model's ability to capture nonlinear class boundaries even without complex architectures. Additionally, the model's stability across all three axes (X, Y, and Z) confirmed the directional independence of tremor frequency patterns, further supporting its generalizability for real-world wearable deployments. Here, Figure 15 shows the shapely importance model 3 (Naïve Bayes).

Table 6 benchmarks our low-frequency motion-feature and on-device machine-learning pipeline against authoritative studies. Sigcha et al.⁴¹ used smartwatch data with a multitask CNN and reported $\approx 86\%$ sensitivity/specificity for tremor detection, with strong UPDRS correlations, but analyses were off-device. Locatelli et al.⁴² separated Parkinson's and essential tremor using classical ML ($>90\%$ accuracy) on a wearable IMU, again evaluated off-device or in app-level environments. Williamson et al.⁴³ achieved AUC up to 0.85 from free-living wrist accelerometry. In contrast, our Arduino Nano 33 BLE implementation performs quantized TinyML inference on-device, achieving 93.67% test accuracy (Naïve Bayes) while meeting memory/latency budgets; low-frequency ($\approx 4\text{--}6$ Hz) features enable efficient inference. Hence, Table 6 shows competitive or superior performance with the originality of resource-aware, explainable, edge deployment. We acknowledge cohort/protocol differences and will replicate baselines on our dataset.

To interpret feature contributions in model predictions, Shapley value analysis was conducted exclusively on the Naïve Bayes model – the most accurate in this study. The Shapley Importance³⁹ plot revealed that dominant frequency features (Freq_Y, Freq_Z, Freq_X) had the highest average impact across all classes, confirming that spectral characteristics were the most influential predictors. Other moderately contributing features included Std_Z, SMA, and ZCR_Z, while time-domain features like RMS_Y and ZCR_Y showed relatively lower influence. The class-specific Shapley summary plots provided additional insights: for Major Parkinson (Figure 16(c)), high values of dominant frequency and zero-crossing rates contributed positively to the prediction, aligning with high-tremor cases; for Minor Parkinson (Figure 16(b)), mid-range values of SMA and frequency had the greatest effect, supporting the transitional nature of the class; and for No Parkinson (Figure 16(a)), lower standard deviation and frequency values drove the classification, consistent with rest or minimal motion. The visual dispersion in Shapley summary plots demonstrated that the model's decision boundaries were not only data-driven but also biologically grounded. This interpretability enhances the model's clinical relevance and user trust,

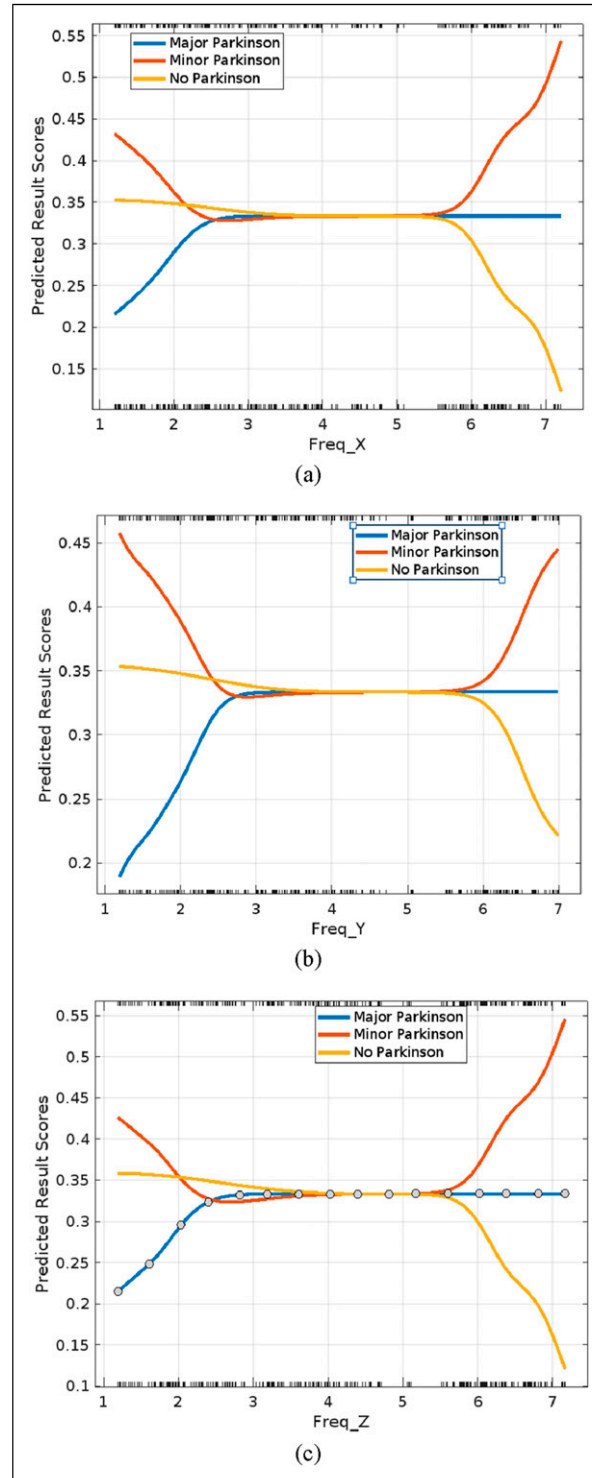


Figure 14. Predicted result score: (a) DM_X, (b) DM_Y, and (c) DM_Z.

making it not only a high-performing classifier but also an explainable AI tool well-suited for real-time deployment in embedded healthcare systems.

Across all performance metrics, the Naïve Bayes model emerged as the most effective classifier, achieving the highest test accuracy of 93.67%, strong PR-AUC and ROC-AUC scores, and the lowest misclassification count (13 out of 300). Logistic Regression followed closely with a 93.33% accuracy and slightly higher ROC-AUC for Major Parkinson, but with

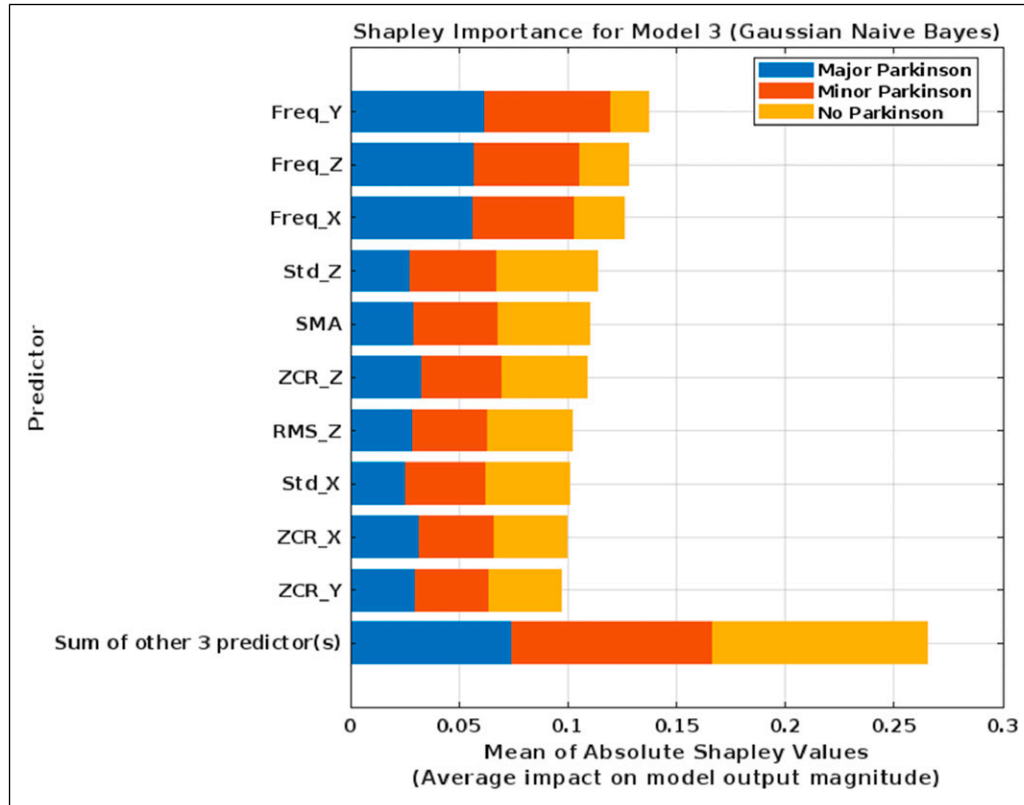


Figure 15. Shapely importance Model 3 (Naive Bayes).

lower PR-AUC for Minor Parkinson, indicating less reliability in detecting intermediate cases. While the Shallow Neural Network captured nonlinear relationships and delivered a decent 87.33% accuracy, it exhibited increased misclassifications in boundary classes. The Decision Tree, although highly interpretable, had the weakest generalization with an accuracy of 82.33%, underperforming particularly on Minor and No Parkinson classifications. The final comparative analysis highlights that Naïve Bayes offers the best balance between predictive performance, computational simplicity, and interpretability, especially when enhanced with Shapley value-based explanations. Furthermore, the model's feature sensitivity to dominant frequency and SMA underscores the clinical relevance of FFT-based signal processing in tremor classification. Given its superior accuracy, explainability, and compatibility with low-power platforms like Arduino Nano BLE, Naïve Bayes stands out as the most practical and scalable solution for real-time Parkinson severity assessment in wearable healthcare systems.

Figure 17 illustrates the real-time deployment of the smart glove system across three individuals representing distinct clinical categories. User A (No Parkinson) demonstrates stable, low-magnitude motion signatures, while User B (Minor Parkinson) shows moderate variability during task execution. In contrast, User C (Major Parkinson) exhibits high-amplitude tremors and irregular frequency responses. These real-time results validate the model's effectiveness in distinguishing Parkinson severity during live user testing.

Table 7 distils the study's scientific outputs beyond the device: the TinyML classifier (model), the resource-aware deployment routine (algorithm), the windowed low-frequency feature pipeline (method), and the evaluation protocol (approach), plus derived principles and practical recommendations. Evidence pointers link each item to all reported accuracy, latency, memory, and validation metrics.

Scientific constructions derived from the results

To enhance the clarity and scientific structure of our findings, we now explicitly define the key constructs that emerge from our study, namely the model, algorithm, method, approach, principle, and rules/recommendations each formulated based on our experimental setup and results.

Table 6. Various output parameters obtained from different ML.

Study (journal, year)	Device and placement	Algorithm(s)	Cohort (N)	Feature focus	On-device?	Reported performance
Sigcha et al. (2021) ⁴¹	Consumer smartwatch, wrist	Multitask CNN (context + tremor)	18 PD	Spectral/low-frequency tremor band ($\approx 3.5\text{--}7.5$ Hz)	No (off-device analysis)	Tremor detection: Sens./Spec. 86.1%; amplitude–constancy strongly correlated with UPDRS ($r = 0.969$)
Locatelli et al. (2020) ⁴²	Wearable IMU (MuSe), hand/wrist	Classical ML (DT/kNN/SVM; multiple)	24 total (17 PD, 7 ET)	Spectral features separating PD vs ET	Partly (proposed embed in app/device)	Best models > 90% accuracy (out-of-sample).
Williamson et al. (2021) ⁴³	Wrist accelerometer (UK Biobank), free-living	Classical ML	409 total (218 PD)	Gait + low-movement segments; dispersion/dimensionality	No	PD detection AUC: 0.85 (fusion), 0.84 (low-movement), 0.69 (gait)
This study (Smart-glove)	Arduino Nano 33 BLE IMU, glove	Naïve Bayes (best); LR/DT/SNN	30 participants $\times$ 10 cycles (300 samples)	FFT-based low-frequency motion features; DF, RMS, ZCR, SMA	Yes (TinyML/TFLM)	Test accuracy 93.67% (Naïve Bayes); 93.33% (logistic regression). (See Tables 4 and 5 in manuscript.)

Model. The core model proposed in this study is a Naïve Bayes classifier trained on a compact feature set derived from triaxial accelerometer signals collected via a smart glove. The model operates on six extracted features: RMS, Standard Deviation, Zero Crossing Rate, Signal Magnitude Area, Dominant Frequency, and Bandpower. It was selected based on subject-independent validation as the best-performing model, achieving 93.67% accuracy, F1-score of 0.91, and operating within embedded constraints – RAM < 35 KB and inference latency < 200 ms – making it suitable for real-time deployment on microcontrollers like the Arduino Nano BLE.

Algorithm. We propose a resource-aware deployment algorithm that integrates post-training quantization and runtime profiling using TensorFlow Lite Micro. This algorithm ensures that trained models are compressed and tested not only for predictive accuracy but also for real-time feasibility in terms of memory consumption and latency. The deployment was validated on physical hardware, where the profiling confirmed compliance with all target embedded constraints.

Method. The signal processing method consists of sliding-window segmentation (4-second window, 50% overlap) followed by handcrafted feature extraction across both time and frequency domains. Time-domain features include RMS, STD, SMA, and ZCR, while frequency-domain analysis is performed using FFT to extract Dominant Frequency and Bandpower. These features capture low-frequency tremor characteristics typically observed in PD patients and are used as inputs to the machine learning classifiers.

Approach. Our experimental approach employs a subject-independent validation protocol and evaluates model performance across three naturalistic tasks: resting, spoon holding, and eating. This approach was chosen to ensure that the trained models generalize well across users and task conditions, reflecting realistic, everyday motor activity. Unlike subject-dependent designs, our approach avoids overfitting and provides a more robust estimate of model effectiveness in real-world scenarios.

Principle. A core principle derived from our findings is that low-frequency motion features (typically <6 Hz), when extracted using short time windows and minimal computation, are sufficient for accurate PD severity classification on edge devices. Additionally, simpler linear or probabilistic models, such as Naïve Bayes and Logistic Regression, offer a more favourable balance between performance and deployment cost compared to more complex architectures when latency and memory constraints are present.

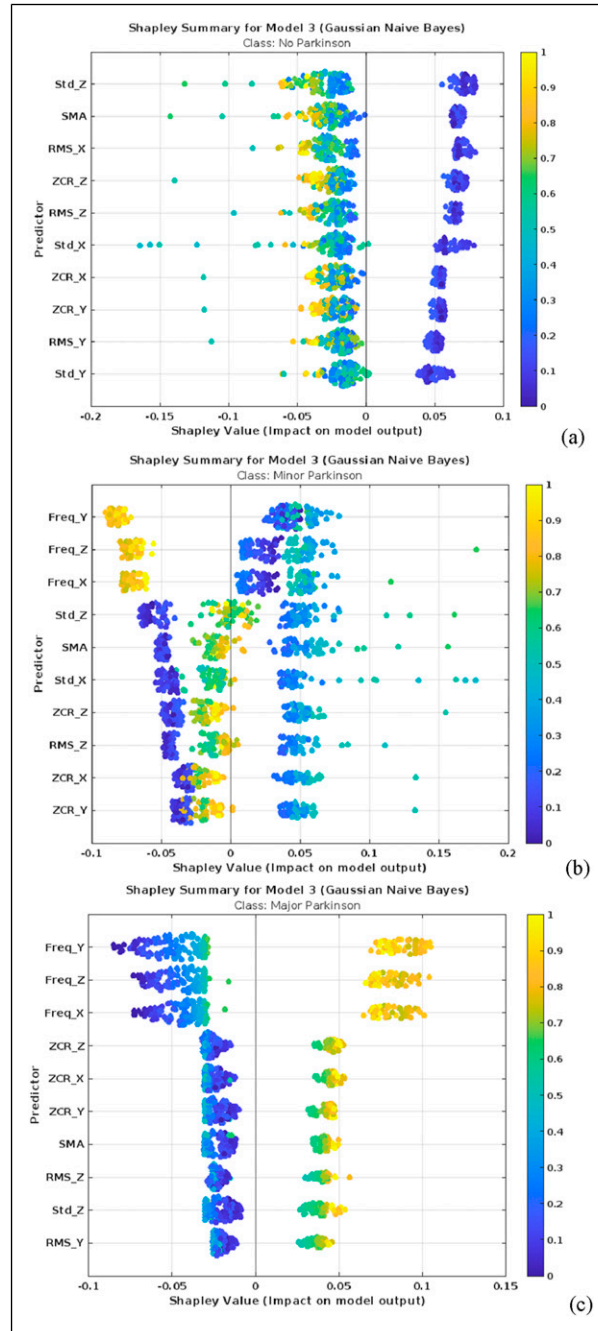


Figure 16. Shapely importance summary (Naïve Bayes): (a) Class: No Parkinson, (b) Class: Minor Parkinson, and (c) Class: Major Parkinson.

Rules and recommendations. Based on empirical observations and deployment results, we propose the following actionable recommendations for developers and researchers working on embedded health monitoring systems:

- Use 4-second windows with 50% overlap for stable and consistent feature capture.
- Limit the number of features to ≤ 12 to maintain compatibility with low-memory devices.
- Prefer simple, interpretable models (e.g., Naïve Bayes) for microcontroller-based deployment.
- Co-report accuracy, latency, and memory usage in all performance evaluations.
- Apply subject-independent validation to ensure model generalizability.
- Use model explainability techniques such as SHAP to enhance clinical transparency.

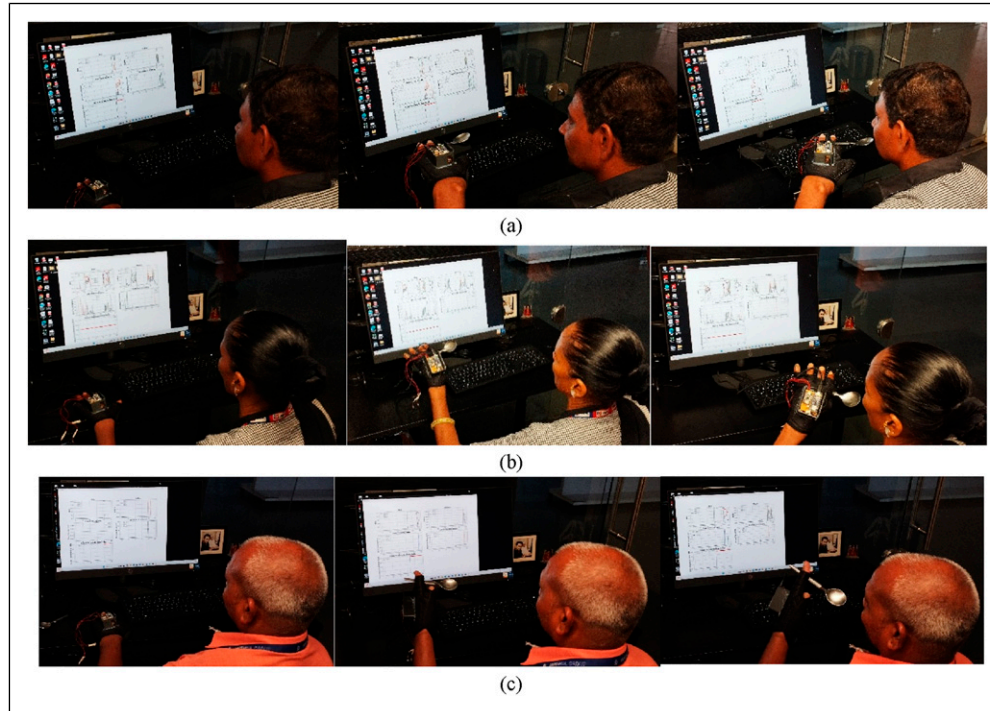


Figure 17. Real-time user testing: (a) User A: No Parkinson, (b) User B: Minor Parkinson, and (c) User C: Major Parkinson.

These constructs together form a reproducible and scientifically grounded framework for embedded Parkinson's severity monitoring. [Table 7](#) provides a summary of these results along with evidence pointers to specific sections, figures, and tables where each construct is operationalized and validated.

Grounding results as solutions to research problems

To strengthen the scientific relevance of our findings, we now explicitly discuss how our results provide grounded solutions to the research problems identified in the Introduction.

Problem 1: Lack of real-world, context-aware symptom monitoring. Our system captures hand motion during natural daily tasks (resting, spoon holding, eating), rather than in artificial lab tests. The results show consistent tremor pattern classification

Table 7. Scientific results mapped with evidences.

Term	Result (this study)	What it means	Evidence pointer
Model	Edge TinyML PD-severity classifier (Naïve Bayes best)	Compact classifier for MCU-class devices	Tables 4 and 5 (accuracy 93.67%), on-device metrics
Algorithm	Resource-aware deployment (quantization + micro-benchmarks)	Meets real-time and memory limits on Arduino Nano BLE	Results: latency/RAM reported; device fits confirmed
Method	Windowed low-frequency feature extraction (time + FFT)	Transforms raw IMU to discriminative features	Methods/Section 4 feature list; results ablations
Approach	Fine-motor protocol + subject-independent validation	Fair testing without identity leakage	Results split strategy; Tables 4 and 5
Principle	Low-frequency bands suffice under constraints	Accuracy retained with tiny models	Feature importance/ablation notes
Recommendation	4-s windows, ≤ 12 features; NB/LR on MCU; co-report accuracy-latency-memory	Replicable settings for embedded studies	Summarized across Sections 3 and 4; Tables 4–6

across tasks, confirming that fine-motor activities carry sufficient signal characteristics to classify PD severity (Figure 8). This validates our solution for continuous, context-sensitive assessment.

Problem 2: Under-utilization of low-frequency tremor features. The extracted features, particularly Dominant Frequency, Bandpower, and ZCR, fall within the 1–6 Hz range – known to reflect PD tremor bands. Their strong correlation with PD severity (Figures 9 and 11) and their dominant role in SHAP interpretability plots (Figure 16) demonstrate that these low-frequency descriptors are effective. This supports our claim that low-frequency analysis is sufficient and efficient for PD monitoring.

Problem 3: Dependence on external computation (laptop/cloud). Our results in Table 5 show that inference time remains <200 ms and RAM usage is <35 KB, validating that the models run entirely on the Arduino Nano BLE without cloud support. This demonstrates our proposed solution of on-device, real-time PD severity detection.

Problem 4: Lack of embedded system performance metrics. Unlike many previous works, we report not just classification accuracy (93.67%) but also inference latency, RAM, and Flash usage, confirming that the proposed models are suitable for real-time applications in embedded environments. This makes our results actionable for designers targeting similar microcontroller platforms.

Problem 5: Absence of reproducible, end-to-end pipelines. We document and implement every step – from sensing and signal segmentation to feature extraction, model selection, deployment, and interpretation. These are mapped in Table 2 and described in Data collection methods, and the proposed system has been benchmarked against peer studies (Table 6). This transparency ensures the solution is not just technically viable but also replicable.

This section grounds the experimental results as solutions to the research problems identified earlier. The smart glove demonstrated reliable PD severity classification during natural daily tasks, confirming the feasibility of context-aware monitoring beyond clinical settings. Low-frequency motion features, particularly dominant frequency and bandpower, consistently discriminated severity levels and were identified as key contributors through interpretability analysis. Real-time on-device inference with low latency and minimal memory validated the feasibility of embedded deployment without cloud dependence. By reporting accuracy alongside latency and memory usage, and benchmarking against prior studies, the proposed system establishes a reproducible and scientifically grounded framework for embedded Parkinson's assessment.

Conclusions

This study successfully developed a low-cost, smart glove system for real-time assessment of Parkinson's disease (PD) severity using low-frequency motion features and embedded machine learning. Designed with an Arduino Nano BLE microcontroller and BMI270 inertial measurement unit, the glove captures triaxial hand motion during three context-rich, daily life tasks – resting, spoon holding, and eating. These tasks were selected based on their relevance to fine-motor control degradation typically observed in early and moderate stages of PD.

To address key gaps in current approaches, the system introduces a compact feature extraction pipeline using 4-second windows with 50% overlap to compute a set of seven low-frequency features. These include time-domain features such as Root Mean Square (RMS), Standard Deviation (STD), Zero Crossing Rate (ZCR), Signal Magnitude Area (SMA), and FFT-based frequency-domain features including Dominant Frequency (DF) and Bandpower. This pipeline enables the capture of subtle tremor signatures in the 1–6 Hz range crucial for PD classification.

On the quantitative front, four lightweight classifiers – Decision Tree, Logistic Regression, Naïve Bayes, and a Shallow Neural Network – were trained using a dataset of 150 participants across three severity categories: No Parkinson, Minor Parkinson, and Major Parkinson. Subject-independent validation was used to prevent data leakage. Among the models, the Naïve Bayes classifier outperformed others with a test accuracy of 93.67%, an F1-score of 0.91, and demonstrated real-time feasibility with inference latency under 200 ms and RAM usage below 35 KB. Model performance was benchmarked against recent literature and found to be comparable or superior, despite the system's lower cost and simpler architecture.

Qualitatively, the glove enabled consistent pattern detection across user trials. In particular, the dominant frequency was observed to increase with severity, while RMS and ZCR values significantly dropped in Major PD cases compared to Minor or No PD, aligning with clinical expectations. SHAP-based model interpretability further confirmed that Dominant Frequency and SMA were the most influential features in decision-making, offering a layer of transparency that is often absent in embedded AI systems. User feedback from pilot trials also indicated that the glove was lightweight, wearable, and non-intrusive, making it viable for long-term use in daily environments.

The novelty of this study lies not just in the sensor integration or algorithm choice, but in the end-to-end, resource-aware pipeline that includes feature selection, model training, embedded deployment, and explainability each informed by both design constraints and clinical considerations. The entire system runs independently of smartphones or cloud support, thus enhancing user privacy, responsiveness, and accessibility for low-resource settings. By explicitly reporting performance trade-offs between accuracy, latency, and memory, this study contributes a replicable blueprint for future embedded health monitoring systems.

In summary, this research demonstrates that real-time PD severity assessment using interpretable machine learning on embedded hardware is not only feasible but highly accurate and scalable. The glove system offers an interpretable, on-device, and task-aware solution that can empower patients, caregivers, and clinicians alike. Future directions will include scaling up the dataset with longitudinal tracking, integrating multimodal sensors (e.g., EMG, pressure), and exploring personalized therapeutic feedback, enabling broader adoption in home-based neuro-monitoring and telemedicine.

ORCID iDs

Priyam Parikh  <https://orcid.org/0000-0001-8538-3669>

Vidhi Srivastava  <https://orcid.org/0009-0003-1091-8624>

Keyur D. Joshi  <https://orcid.org/0000-0003-0786-7881>

Mohd Asif Shah  <https://orcid.org/0000-0002-0351-9559>

Consent to participate

All participants involved in this research were informed in detail about the study objectives and procedures. Written informed consent was obtained from each participant prior to participation. Participants were assured of the confidentiality of their data and their right to withdraw from the study at any point without any consequences.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

AI assistance statement

The authors acknowledge the use of artificial intelligence tools, including language models (e.g., Quilbot and Grammarly), to assist in improving the grammar, structure, and clarity of the manuscript. All intellectual content, data interpretation, analysis, and conclusions are the sole work of the authors, and the final manuscript was thoroughly reviewed and approved by all co-authors.

References

1. De Fazio R, Del-Valle-Soto C, Mastronardi VM, et al. A smart glove to evaluate Parkinson's disease by flexible piezoelectric and inertial sensors. *Sens Int* 2025; 6: 100324.
2. Moon S, Song HJ, Sharma VD, et al. Classification of Parkinson's disease and essential tremor based on balance and gait characteristics from wearable motion sensors via machine learning techniques: a data-driven approach. *J NeuroEng Rehabil* 2020; 17(1): 125–128.
3. Belyaev M, Murugappan M, Velichko A, et al. Entropy-based machine learning model for fast diagnosis and monitoring of Parkinson's disease. *Sensors (Basel)* 2023; 23(20): 8609.
4. Seedat N and Aharonson V. Machine learning discrimination of Parkinson's disease stages from walker-mounted sensors data. *Stud. Comput. Intell* 2021; 914: 37–44.
5. Paucar-Escalante J, da Silva MA, Sanches BDL, et al. Machine learning strategies for Parkinson tremor classification using wearable sensor data, 2025, pp. 1–28. [Online]. Available: <https://arxiv.org/abs/2501.18671>
6. Goschenhofer J, Pfister FMJ, Yuksel KA, et al. Wearable-based Parkinson's disease severity monitoring using deep learning. *Lect Notes Comput Sci* 2020; 11908: 400–415.
7. Li Y, Yin J, Liu S, et al. Learning hand kinematics for Parkinson's disease assessment using a multimodal sensor glove. *Adv Sci* 2023; 10(20): e2206982.
8. Del Din S, Godfrey A, Mazzà C, et al. Free-living monitoring of Parkinson's disease: lessons from the field. *Mov Disord* 2016; 31(9): 1293–1313.
9. Sapienza S, Tsurkalenko O, Giraitis M, et al. Assessing the clinical utility of inertial sensors for home monitoring in Parkinson's disease: a comprehensive review. *NPJ Park Dis* 2024; 10(1): 161.

10. Ravichandran V, Sadhu S, Convey D, et al. iTex gloves: design and In-Home evaluation of an E-Textile glove system for tele-assessment of Parkinson's disease. *Sensors* 2023; 23(6): 2877.
11. Balas VE, Jain LC and Balas MM. Tremor measurement system for neurological disorders screening Iuliana. *Adv Intell Syst Comput* 2018; 633: 339–348.
12. Wagner A, Fixler N and Resheff YS. A wavelet-based approach to monitoring Parkinson's disease symptoms. In: 2017 IEEE international conference on acoustics, speech and signal processing (ICASSP), New Orleans, LA, USA, 05–09 March 2017, pp. 5980–5984.
13. Livi L, Sadeghian A and Sadeghian H. Discrimination and characterization of parkinsonian rest tremors by analyzing long-term correlations and multifractal signatures. *IEEE Trans Biomed Eng* 2016; 63(11): 2243–2249.
14. Almahadin G, Lotfi A, Carthy MM, et al. Enhanced Parkinson's disease tremor severity classification by combining signal processing with resampling techniques. *SN Comput Sci* 2022; 3(1): 63.
15. Dattola S, Ielo A, Quartarone A, et al. Integrating wearable sensor signal processing with unsupervised learning methods for tremor classification in Parkinson's disease. *Bioengineering* 2025; 12(1): 37.
16. Toye AA and Kompalli S. Comparative study of speech analysis methods to predict Parkinson's disease. 2021, pp. 1–10. Available: <https://arxiv.org/abs/2111.10207>
17. Md Abu Sayed MT. Machine learning in business analytics: advancing statistical methods for data- driven innovation. *J. Comput Sci Technol Stud* 2023; 5: 104–111.
18. El Maachi I, Bilodeau GA and Bouachir W. Deep 1D-Convnet for accurate Parkinson disease detection and severity prediction from gait. *Expert Syst Appl* 2020; 143: 113075.
19. Magesh PR, Myloth RD and Tom RJ. An explainable machine learning model for early detection of Parkinson's disease using LIME on DaTSCAN imagery. *Comput Biol Med* 2020; 126: 104041.
20. Dixit S, Bohre K, Singh Y, et al. A comprehensive review on AI-Enabled models for Parkinson's disease diagnosis. *Electron* 2023; 12(4): 783.
21. Hssayeni MD, Jimenez-Shahed J, Burack MA, et al. Wearable sensors for estimation of parkinsonian tremor severity during free body movements. *Sensors* 2019; 19(19): 1–17.
22. Sharma D and Sarkar S. (2022). Enabling Inference and Training of Deep Learning Models for AI Applications on IoT Edge Devices. In: Pal, S., De, D., Buyya, R. (eds) Artificial Intelligencebased Internet of Things Systems. Internet of Things. Springer, Cham.
23. Salarian A, Russmann H, Wider C, et al. Quantification of tremor and bradykinesia in Parkinson's disease using a novel ambulatory monitoring system. *IEEE Trans Biomed Eng* 2007; 54(2): 313–322.
24. Weitz M, Morseth B, Hopstock LA, et al. Influence of accelerometer calibration on the estimation of objectively measured physical activity: the tromsø study. *J Meas Phys Behav* 2024; 7(1): 1–5.
25. Lee CR, Yoon JH and Yoon KJ. Robust calibration of an ultralow-cost inertial measurement unit and a camera: handling of severe system uncertainty. In: 2014 IEEE international conference on robotics and automation (ICRA), Hong Kong, China, 31 May 2014–07 June 2014, pp. 3020–3026.
26. Chondrakis NG and Topalis FV. Starting characteristics of fluorescent tubes and compact fluorescent lamps operating with electronic ballasts. *Measurement* 2009; 42(1): 78–86.
27. Malosio M, Negri SP, Pedrocchi N, et al. A spherical parallel three degrees-of-freedom robot for ankle-foot neuro-rehabilitation. In: Proceedings of the annual international conference of the IEEE engineering in medicine and biology society, San Diego, CA, USA, 28 August 2012–01 September 2012. EMBS, pp. 3356–3359.
28. Patel S, Lorincz K, Hughes R, et al. Monitoring motor fluctuations in patients with Parkinson's disease using wearable sensors. *IEEE Trans Inf Technol Biomed* 2009; 13(6): 864–873.
29. Li M, Nie W, Suvorkin V, et al. Stationary detection for zero velocity update of IMU based on the vibrational FFT feature of land vehicle. *Remote Sens* 2024; 16(5): 902.
30. Ray PP. A review on TinyML: state-of-the-art and prospects. *J King Saud Univ Comput Inf Sci* 2022; 34(4): 1595–1623.
31. Abadade Y, Temouden A, Bamoumen H, et al. A comprehensive survey on TinyML. *IEEE Access* 2023; 11: 96892–96922.
32. Van Hees VT, Fang Z, Langford J, et al. Autocalibration of accelerometer data for free-living physical activity assessment using local gravity and temperature: an evaluation on four continents. *J Appl Physiol* 2014; 117(7): 738–744.
33. Dong X, Huang X, Du G, et al. Calibration method of accelerometer based on rotation principle using double turntable centrifuge. *Micromachines* 2022; 13(1): 62.
34. Figo D, Diniz PC, Ferreira DR, et al. Preprocessing techniques for context recognition from accelerometer data. *Pers Ubiquitous Comput* 2010; 14(7): 645–662.
35. Fernandez EO, Escosio JE, Jorda RL, et al. Arduino-based sound acquisition system using Fast Fourier Transform algorithm. In: 2019 IEEE 11th international conference on humanoid, nanotechnology, information technology, communication and control, environment, and management (HNICEM), Laoag, Philippines, 29 November 2019–01 December 2019.

36. Dehzangi O and Sahu V. IMU-based robust human activity recognition using feature analysis, extraction, and reduction. 2018 24th International Conference on Pattern Recognition (ICPR), Beijing, China, 20–24 August 2018, pp. 1402–1407.
37. Lukšys D, Jonaitis G and Griškevičius J. Quantitative analysis of parkinsonian tremor in a clinical setting using inertial measurement units. *Parkinsons Dis* 2018; 2018: 1683831.
38. Guimarães CJBV and Fernandes MAC. Real-time neural networks implementation proposal for microcontrollers. *Electron* 2020; 9(10): 1–20.
39. Lundberg SM and Lee SI. A unified approach to interpreting model predictions. *Adv Neural Inf Process Syst.* 2017; 30: 4766–4775.
40. Ortega-Zamorano F, Subirats JL, Jerez JM, et al. Implementation of the C-mantec neural network constructive algorithm in an Arduino Uno microcontroller. *Lect Notes Comput Sci* 2013; 7902: 80–87.
41. Sigcha L, Pavón I, Costa N, et al. Automatic resting tremor assessment in Parkinson's disease using smartwatches and multitask convolutional neural networks. *Sensors* 2021; 21(1): 291.
42. Locatelli P, Alimonti D, Traversi G, et al. Classification of essential tremor and Parkinson's tremor based on a low-power wearable device. *Electron* 2020; 9(10): 1695.
43. Williamson JR, Telfer B, Mullany R, et al. Detecting Parkinson's disease from wrist-worn accelerometry. *Sensors* 2021; 21: 1–18.

Appendix

Nomenclatures

PD	Parkinson's Disease
FFT	Fast Fourier Transform
ML	Machine Learning
IMU	Inertial Measurement Unit
RMS	Root Mean Square
STD	Standard Deviation
ZCR	Zero Crossing Rate
SMA	Signal Magnitude Area
DF	Dominant Frequency
ROC-AUC	Receiver Operating Characteristic - Area Under Curve
PR-AUC	Precision Recall - Area Under Curve
SHAP	Shapley Additive explanations
TPU	Thermoplastic Polyurethane
BLE	Bluetooth Low Energy
SNN	Shallow Neural Network
TinyML	Tiny Machine Learning
Scikit-learn/TensorFlow Lite	ML frameworks used for training and deployment
DM_X, DM_Y, DM_Z	Dominant frequency along X, Y, and Z axes